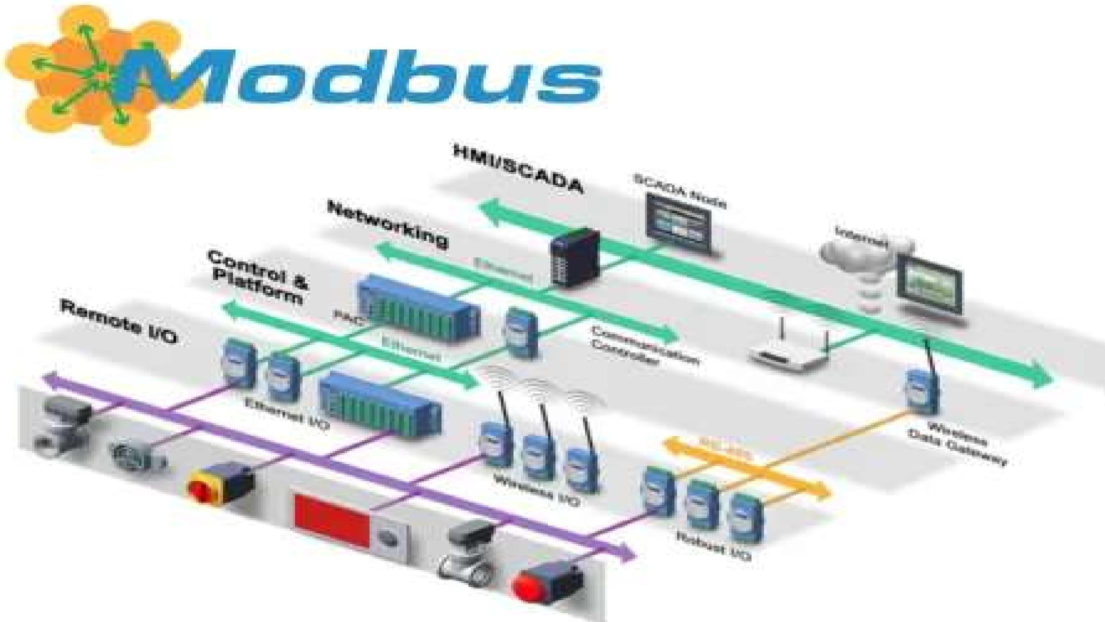
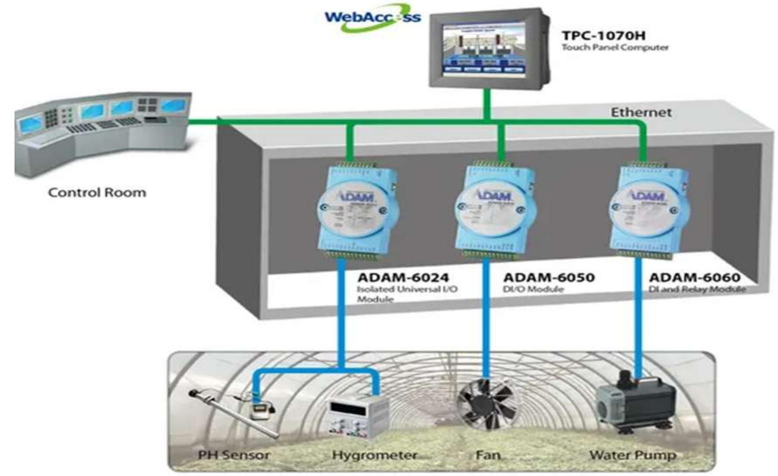
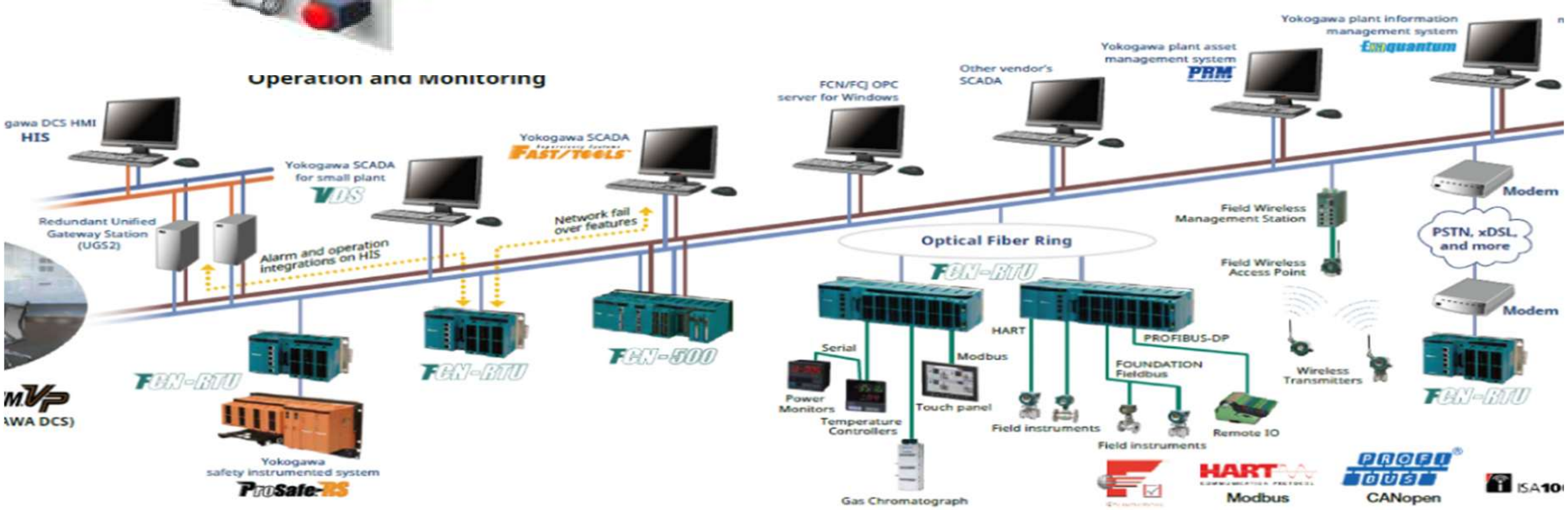
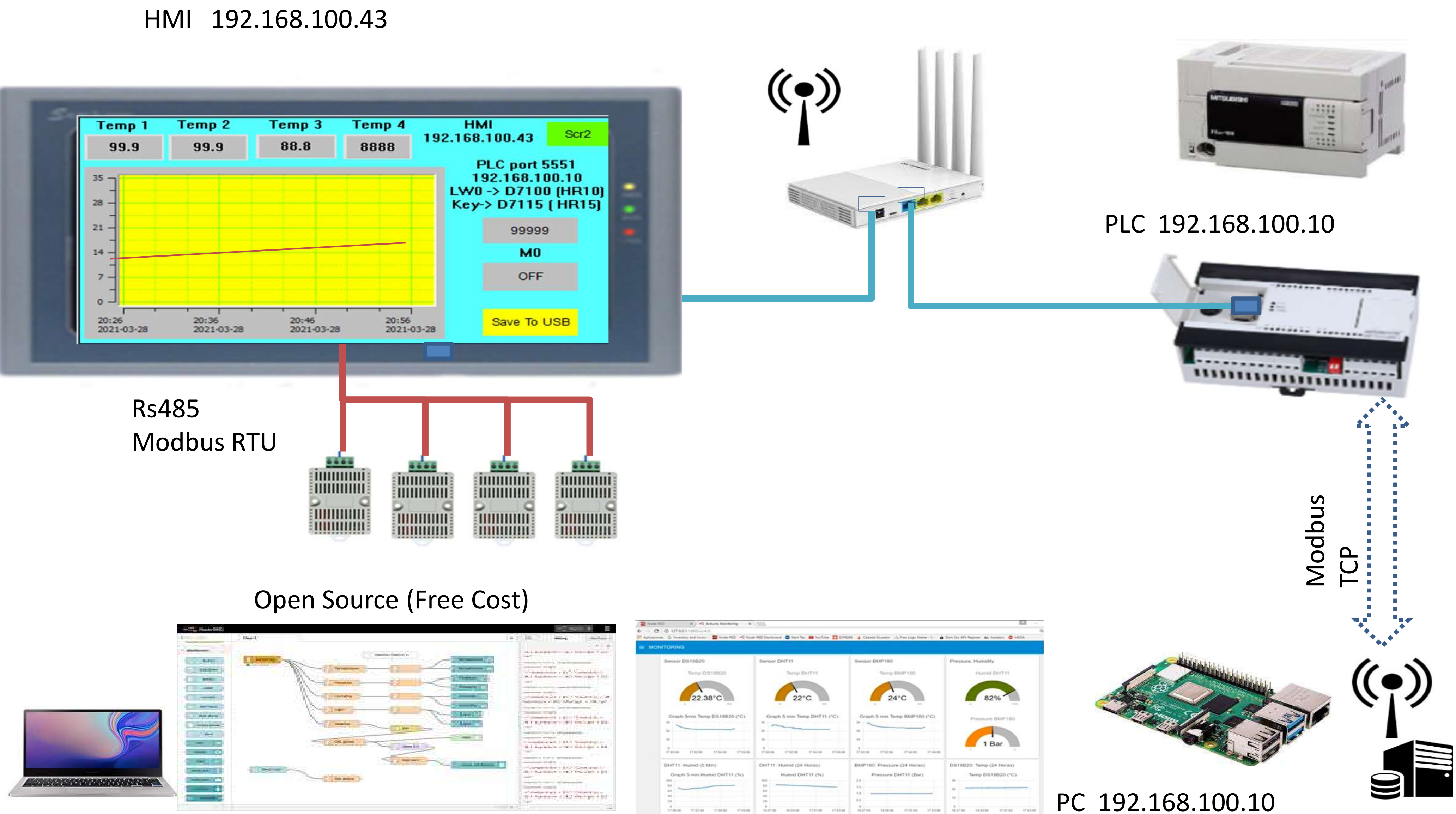
communications and protocols on the control system level

Protocol	Level	Common Applications
ModBus	Device	Manufacturing, Electric Utility
Profibus	Device	Process Industry
DeviceNet	Device	Manufacturing
DNP 3.0	Device	Electric Utility SCADA
BACNet	Control	HVAC Control, Building Automation
ControlNet	Control	Manufacturing
ARCNet	Supervisory	Office Automation, Gaming
Ethernet/IP	Supervisory	Office Automation, Internet

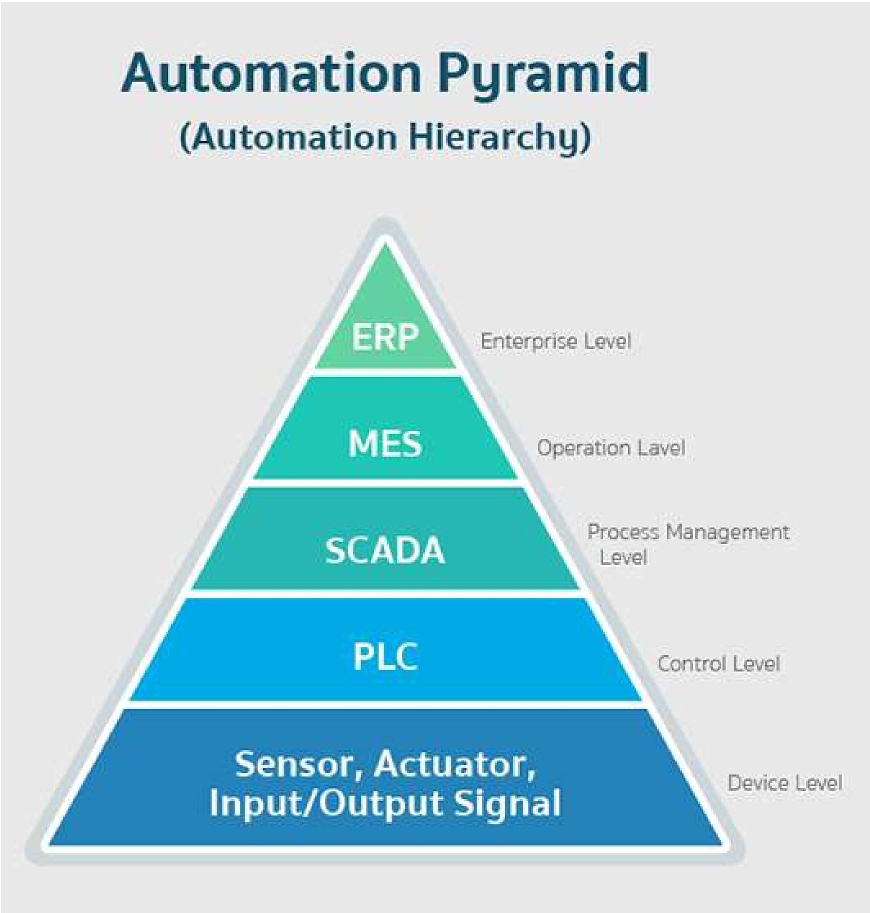

Modbus
HMI/SCADA
SCADA Node
Networking
Internet
Ethernet
Control & Platform
PAC
Communication Controller
Remote I/O
Ethernet I/O
Wireless Data Gateway
Wireless I/O
Robust I/O


WebAccess
TPC-1070H
Touch Panel Computer
Ethernet
Control Room
ADAM-6024
Isolated Universal I/O Module
ADAM-6050
DI/O Module
ADAM-6060
DI and Relay Module
PH Sensor
Hygrometer
Fan
Water Pump


Operation and Monitoring
Yokogawa DCS HMI
HIS
Yokogawa SCADA for small plant
VDS
Yokogawa SCADA
FAST/TOOLS
FCN/FCJ OPC server for Windows
Other vendor's SCADA
Yokogawa plant asset management system
PRM
Yokogawa plant information management system
Exaquantum
Redundant Unified Gateway Station (UGS2)
Alarm and operation integrations on HIS
Network fail over features
Optical Fiber Ring
Field Wireless Management Station
Field Wireless Access Point
Modem
PSTN, xDSL, and more
FCN-RTU
FCN-500
Serial
Modbus
HART
PROFIBUS-DP
FOUNDATION Fieldbus
Power Monitors
Temperature Controllers
Touch panel
Field instruments
Remote IO
Wireless Transmitters
Gas Chromatograph
Yokogawa safety instrumented system
ProSafe-RS
Modbus
CANopen



HMI 192.168.100.43
Temp 1
Temp 2
Temp 3
Temp 4
99.9
99.9
88.8
8888
HMI
192.168.100.43
Scr2
PLC port 5551
192.168.100.10
LW0 -> D7100 (HR10)
Key-> D7115 (HR15)
99999
M0
OFF
Save To USB
20:26
2021-03-28
20:36
2021-03-28
20:46
2021-03-28
20:56
2021-03-28
PLC 192.168.100.10
Rs485
Modbus RTU
Modbus
TCP
Open Source (Free Cost)
PC 192.168.100.10



Protocol	Level	Common Applications
ModBus	Device	Manufacturing, Electric Utility
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DeviceNet	Device	Manufacturing
DNP 3.0	Device	Electric Utility SCADA
BACNet	Control	HVAC Control, Building Automation
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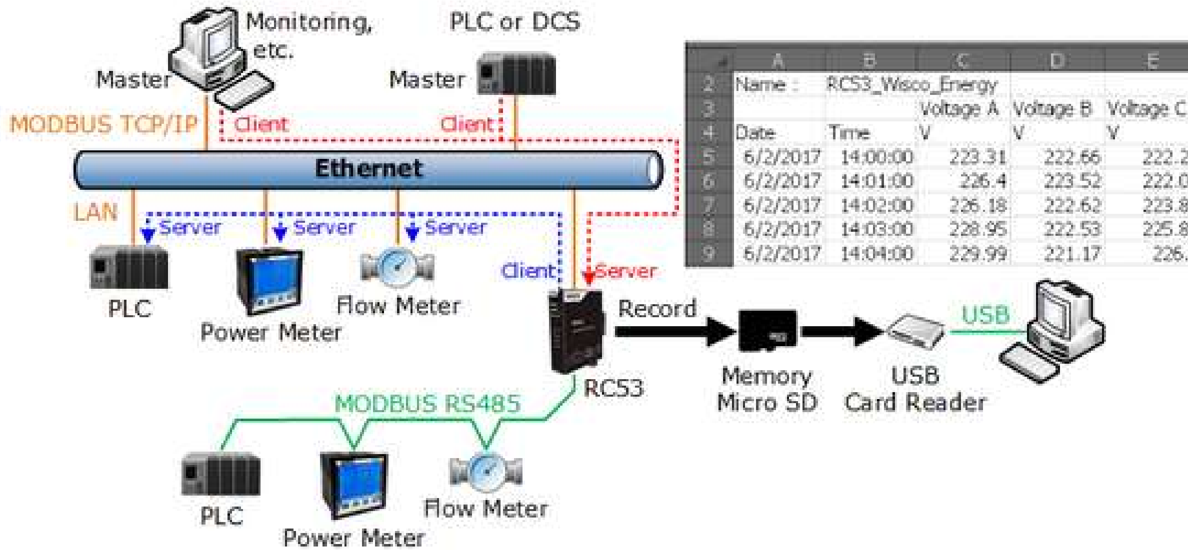
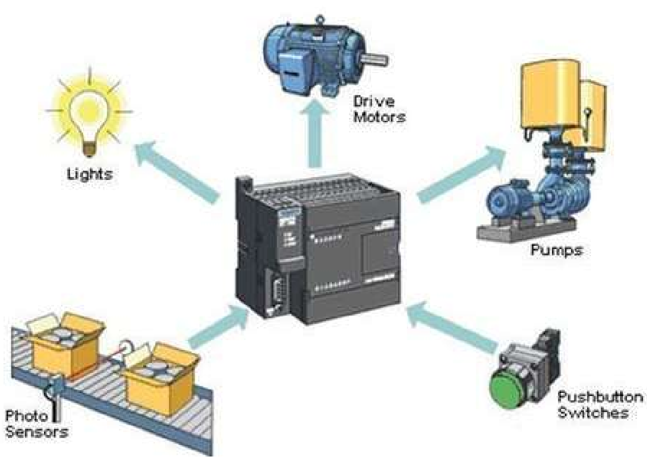
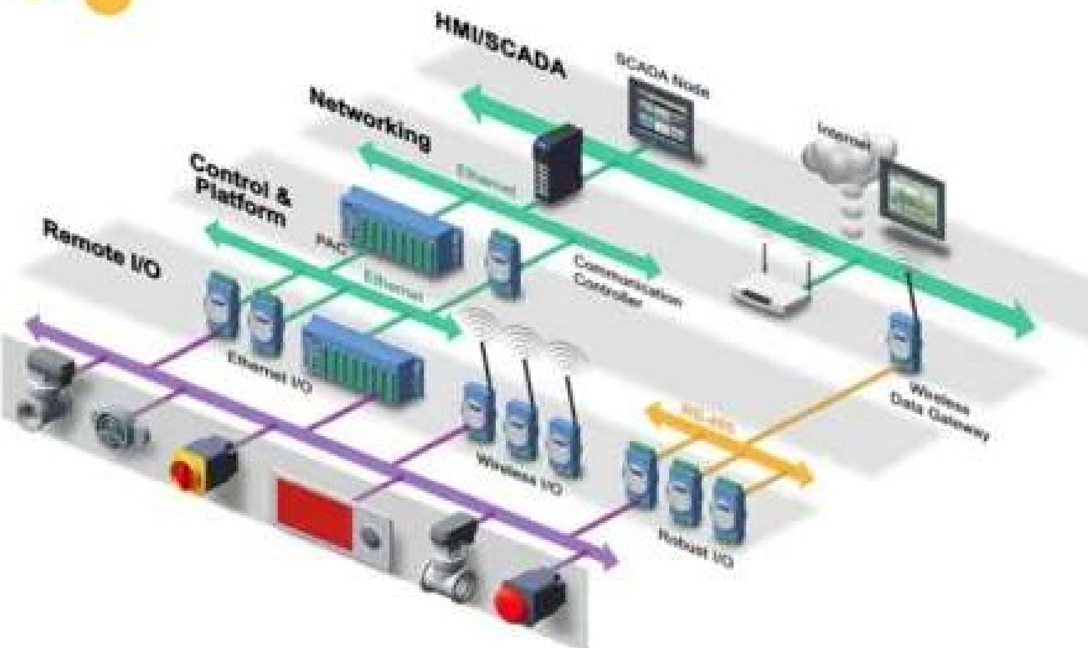
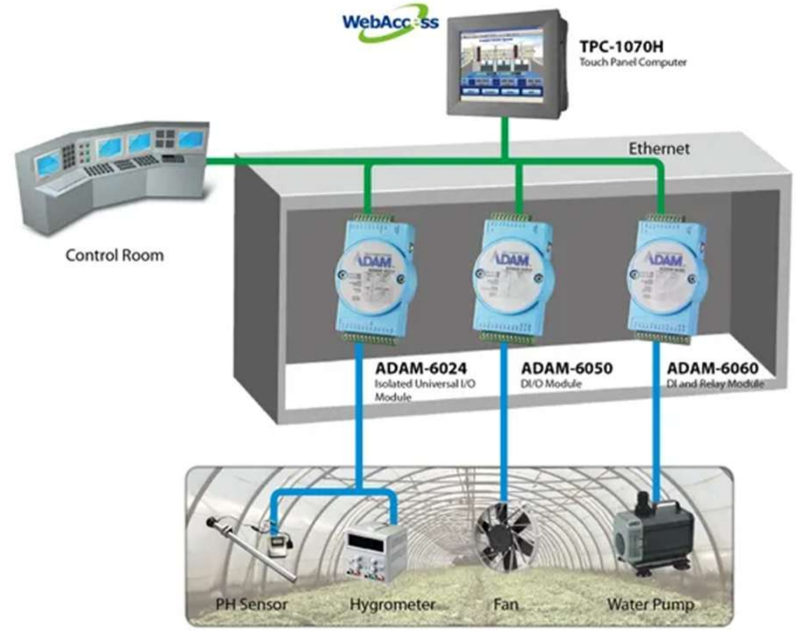
ARCNet	Supervisory	Office Automation, Gaming
Ethernet/IP	Supervisory	Office Automation, Internet

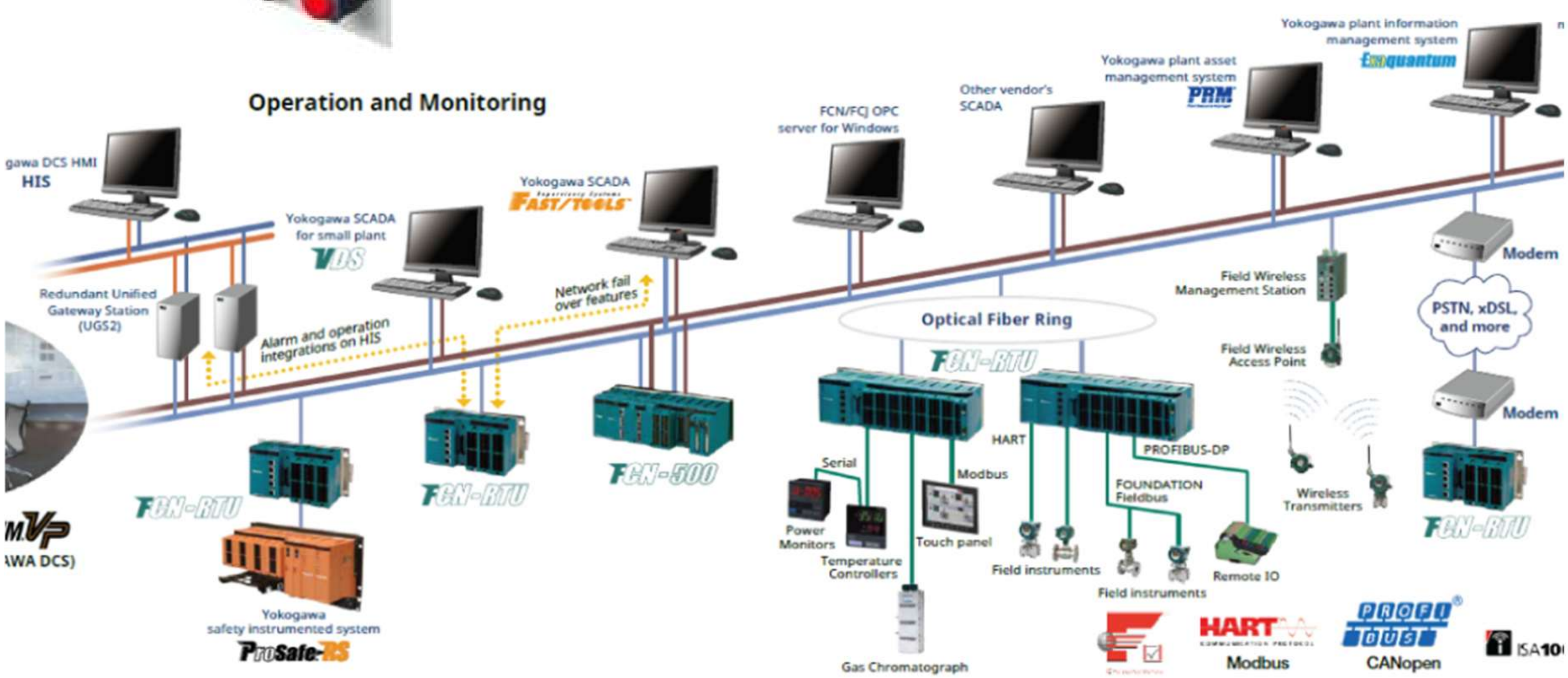




Modbus

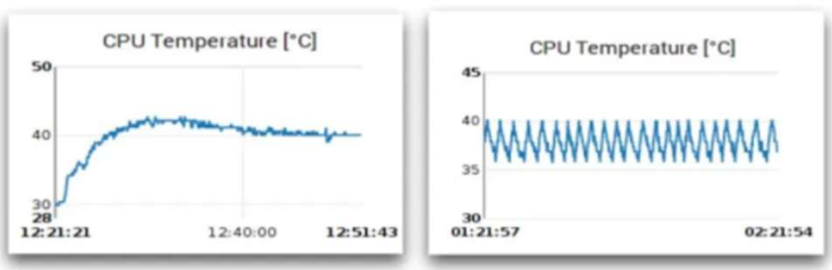

HMI/SCADA
SCADA Node
Networking
Internet
Control &
Platform
PAC
Ethernet
Communication
Controller
Remote I/O
Ethernet I/O
Wireless I/O
Wireless
Data Gateway
Robust I/O

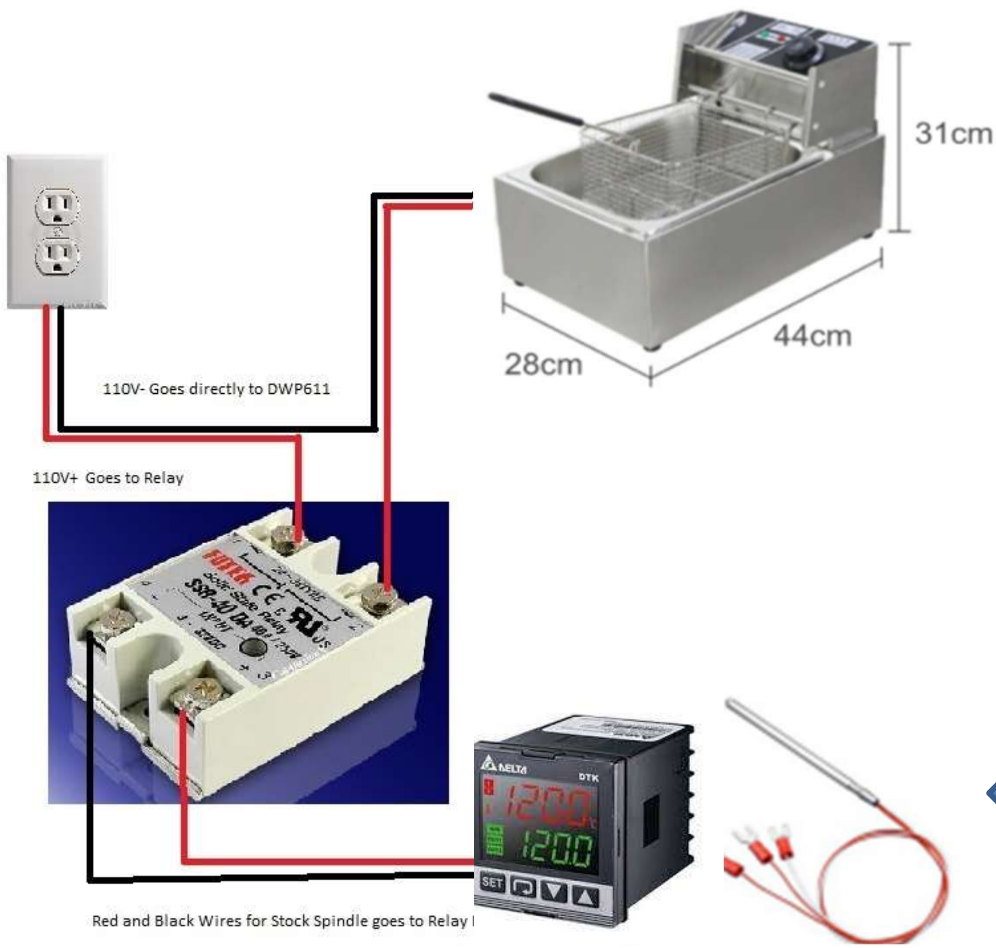

WebAccess
TPC-1070H
Touch Panel Computer
Ethernet
Control Room
ADAM-6024
Isolated Universal I/O
Module
ADAM-6050
DI/O Module
ADAM-6060
DI and Relay Module
PH Sensor
Hygrometer
Fan
Water Pump


Operation and Monitoring
gawa DCS HMI
HIS
Yokogawa SCADA
for small plant
Yokogawa SCADA
FAST/TOOLS
FCN/FCJ OPC
server for Windows
Other vendor's
SCADA
Yokogawa plant asset
management system
PRM
Yokogawa plant information
management system
Redundant Unified
Gateway Station
(UGS2)
Alarm and operation
integrations on HIS
Network fail
over features
Optical Fiber Ring
Field Wireless
Management Station
Field Wireless
Access Point
Modem
PSTN, xDSL,
and more
Modem
FCN-RTU
FCN-RTU
FCN-500
FCN-RTU
FCN-RTU
Serial
HART
PROFIBUS-DP
Modbus
FOUNDATION
Fieldbus
Power
Monitors
Temperature
Controllers
Touch panel
Field instruments
Field instruments
Remote IO
Wireless
Transmitters
AWA DCS)
Yokogawa
safety instrumented system
ProSafe-RS
Gas Chromatograph
HART
Modbus
CANopen

Temperature PID Control

PID control VS On/Off control

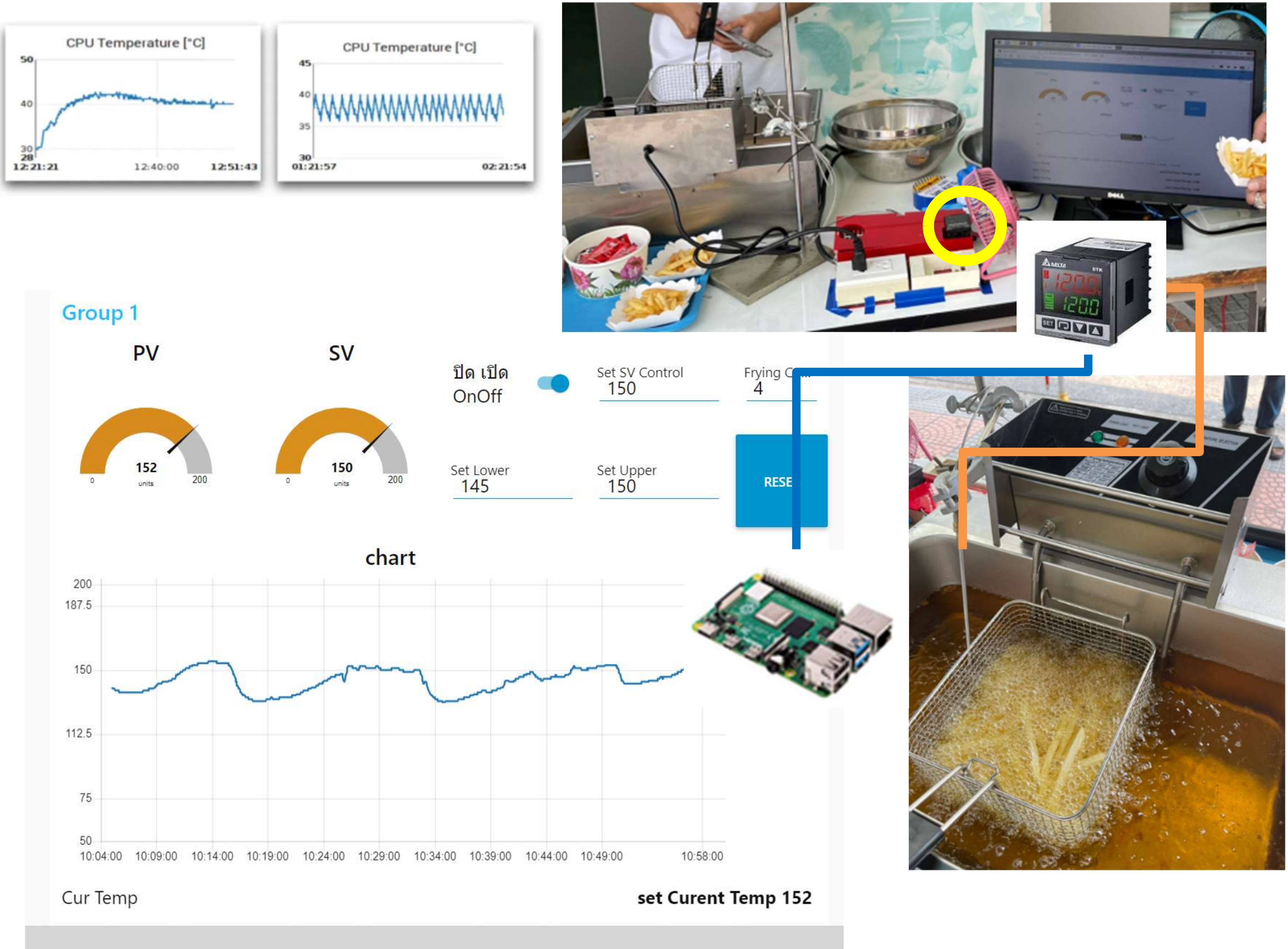




Modbus RTU

DTK4848V12

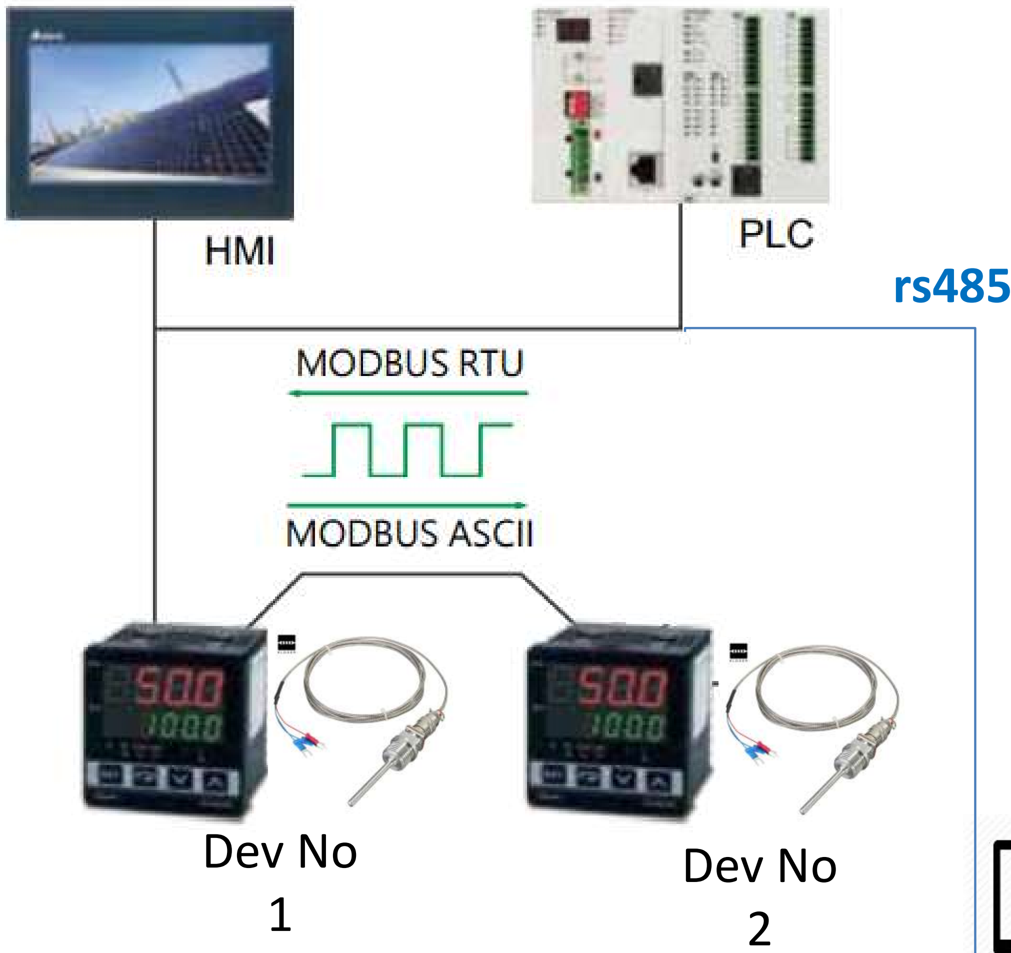
PID control VS On/Off control

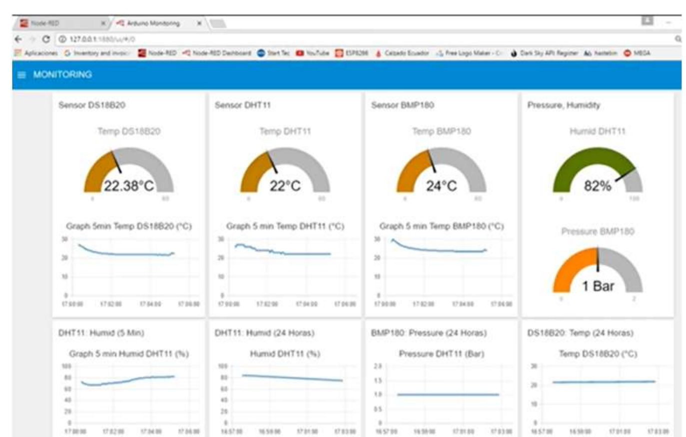


Delta Temperature Controller Model :DTK4848V12

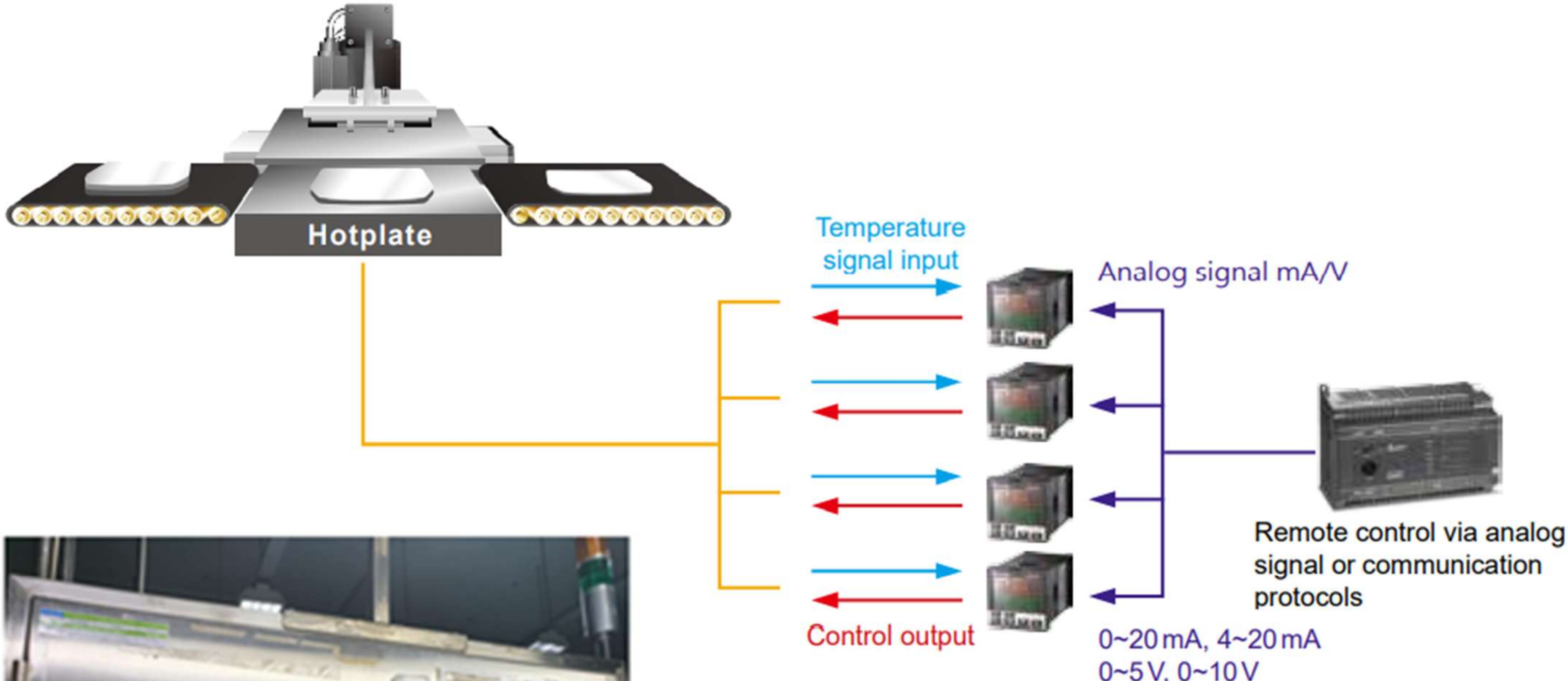
Communication:

- RS-485 communication interface, supporting MODBUS ASCII/RTU communication





Temperature Controller



Point-to-point Control (Proportional Output mA/V)

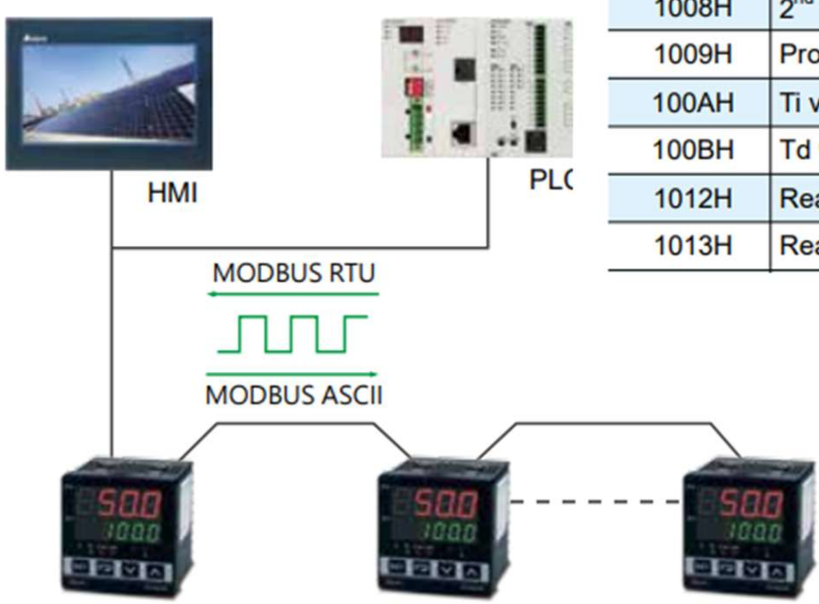
Sets the Present Value by point-to-point control.

Delta Temperature Controller Model :DTK4848V12

RS-485 Communication

DT3 supports baudrate 2,400 to 38,400 bps, MODBUS ASCII/RTU protocol function code 03H and reads maximum 8 words from the register.

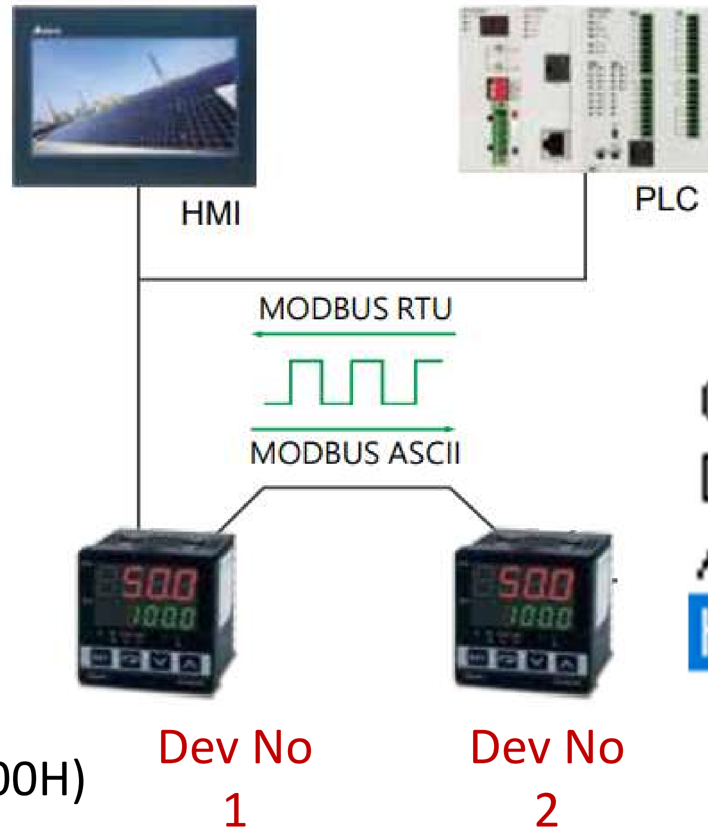
Address	Content	Definition
1000H	Present value (PV)	Measuring unit: 0.1 scale. The following values read mean error occurs. 8002H: Temperature not yet acquired 8003H: Not connected to sensor 8004H: Incorrect sensor
1001H	Set value (SV)	Measuring unit: 0.1 scale
1002H	Upper limit of temp. range	Cannot exceed the default value
1003H	Lower limit of temp. range	Cannot fall below the default value
1005H	Control mode	0: PID, 1: ON/OFF, 2: Manual, 3: FUZZY
1006H	Heating/ Cooling control	0: Heating/Heating, 1: Cooling/Heating, 2: Heating/Cooling, 3: Cooling/Cooling
1007H	1st Heating/ Cooling control cycle	0.1 ~ 99 sec.
1008H	2nd Heating/ Cooling control cycle	0.1 ~ 99 sec.
1009H	Proportional band (PB)	0.1 ~ 999.9
100AH	Ti value	0 ~ 9999
100BH	Td value	0 ~ 9999
1012H	Read/write Output 1 volume	Unit: 0.1%, only valid in manual control mode
1013H	Read/write Output 2 volume	Unit: 0.1%, only valid in manual control mode



Modbus RTU

Communication:

- RS-485 communication interface, supporting MODBUS ASCII/RTU communication



Coil Outputs (000000)
Digital Inputs (100000)
Analog Inputs (300000)
Holding Regs (400000)

400000 (FC 03) + 4096 (1000H)

Address	Content
1000H	Present value (PV)
1001H	Set value (SV)

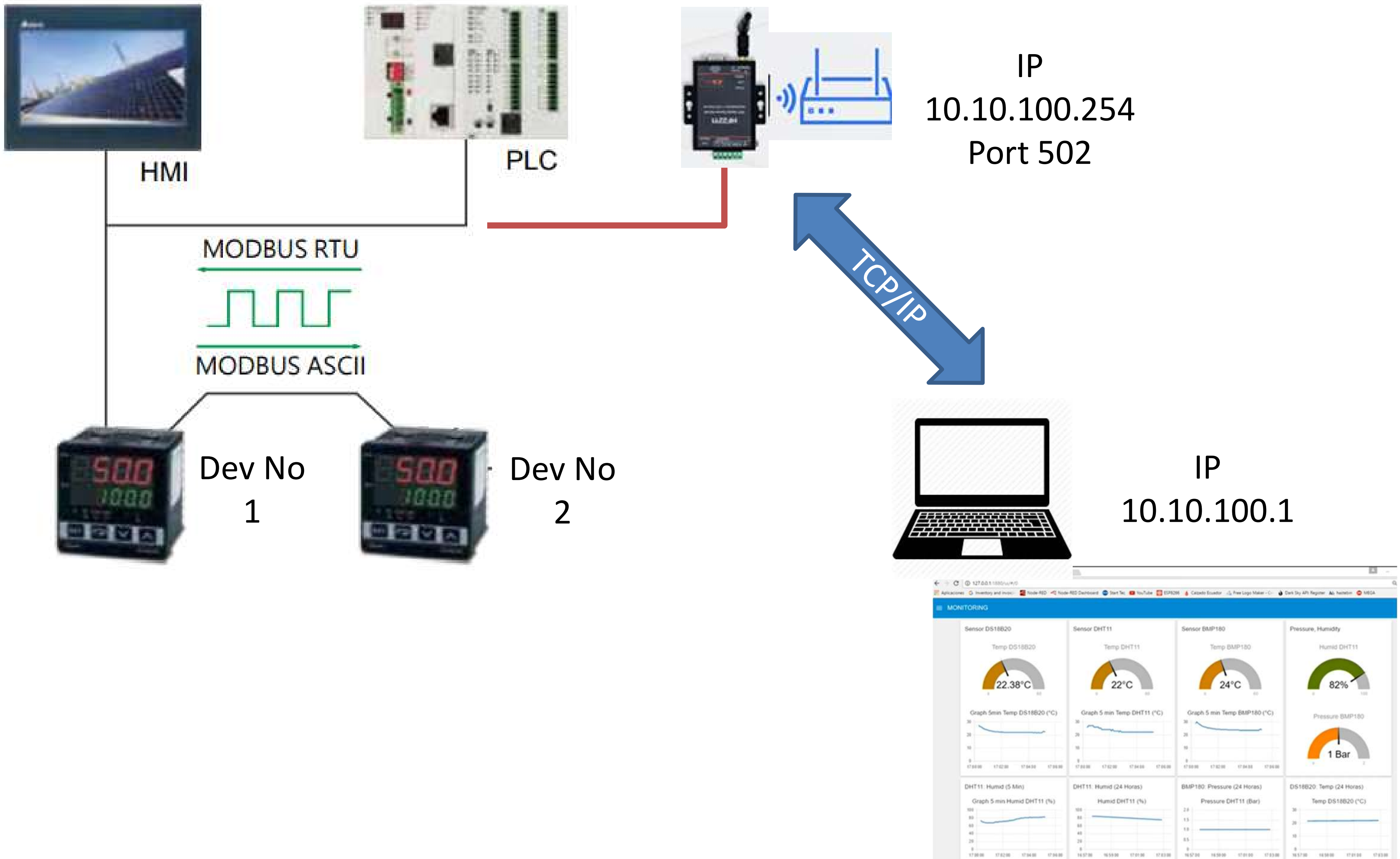
400000 (FC 03) + 4097 (1001H)

Coil Outputs (000000)
Digital Inputs (100000)
Analog Inputs (300000)
Holding Regs (400000)

Address	Content
1000H	Present value (PV)
1001H	Set value (SV)

Communication:

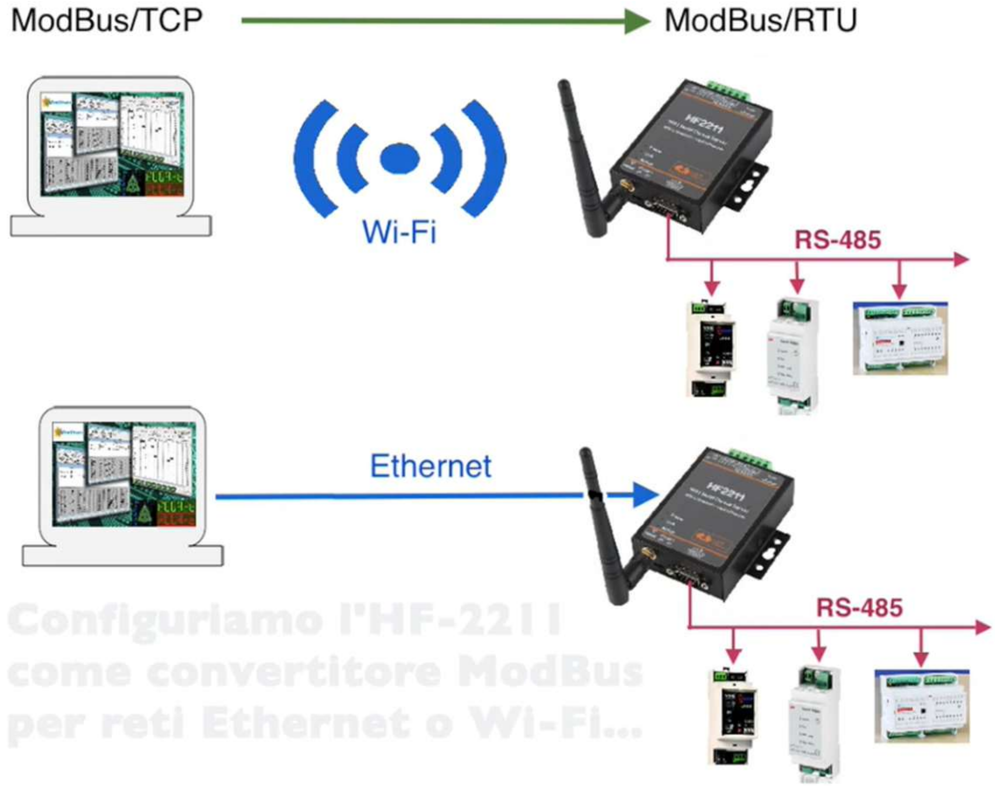
- RS-485 communication interface, supporting MODBUS ASCII/RTU communication

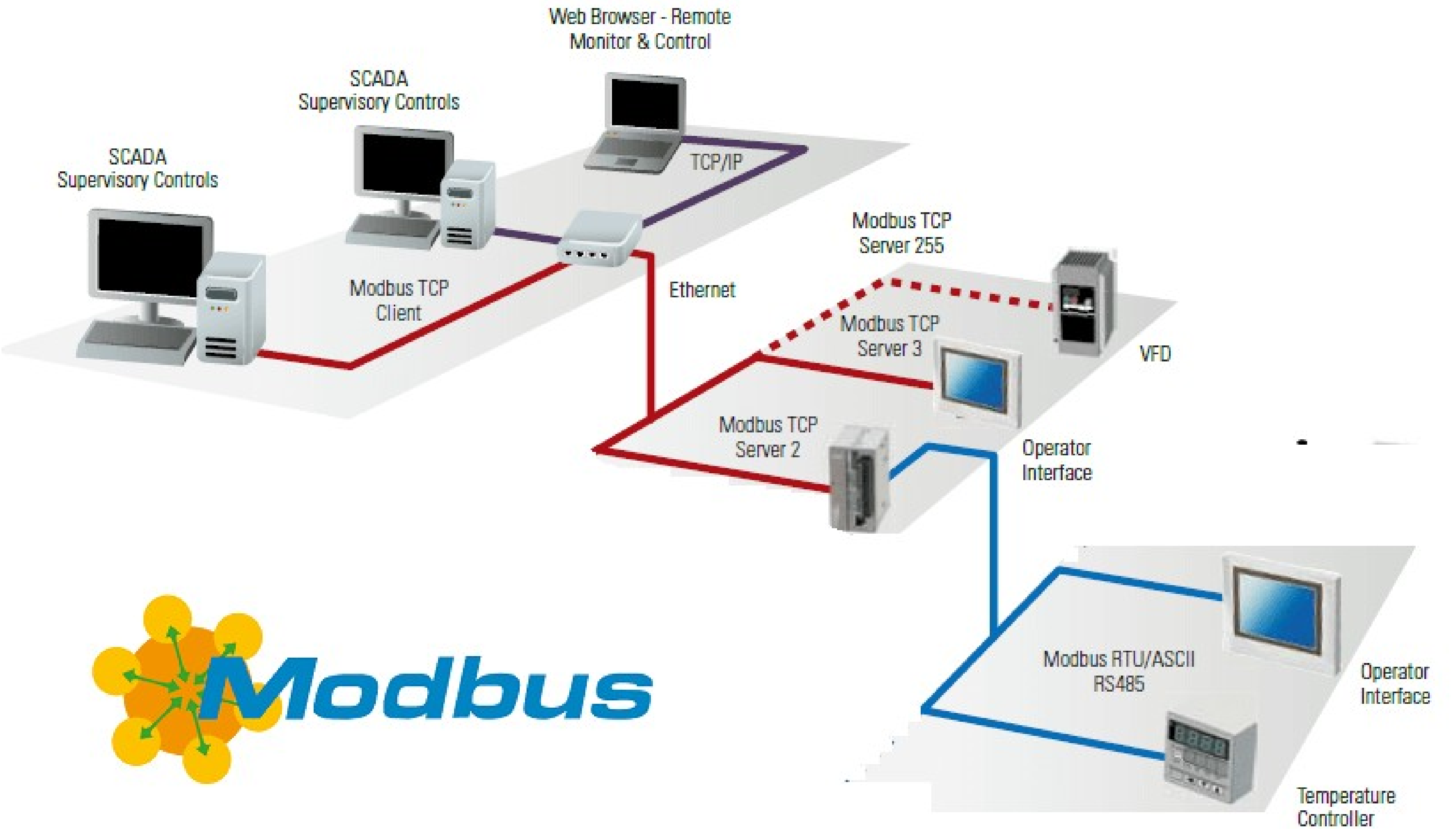


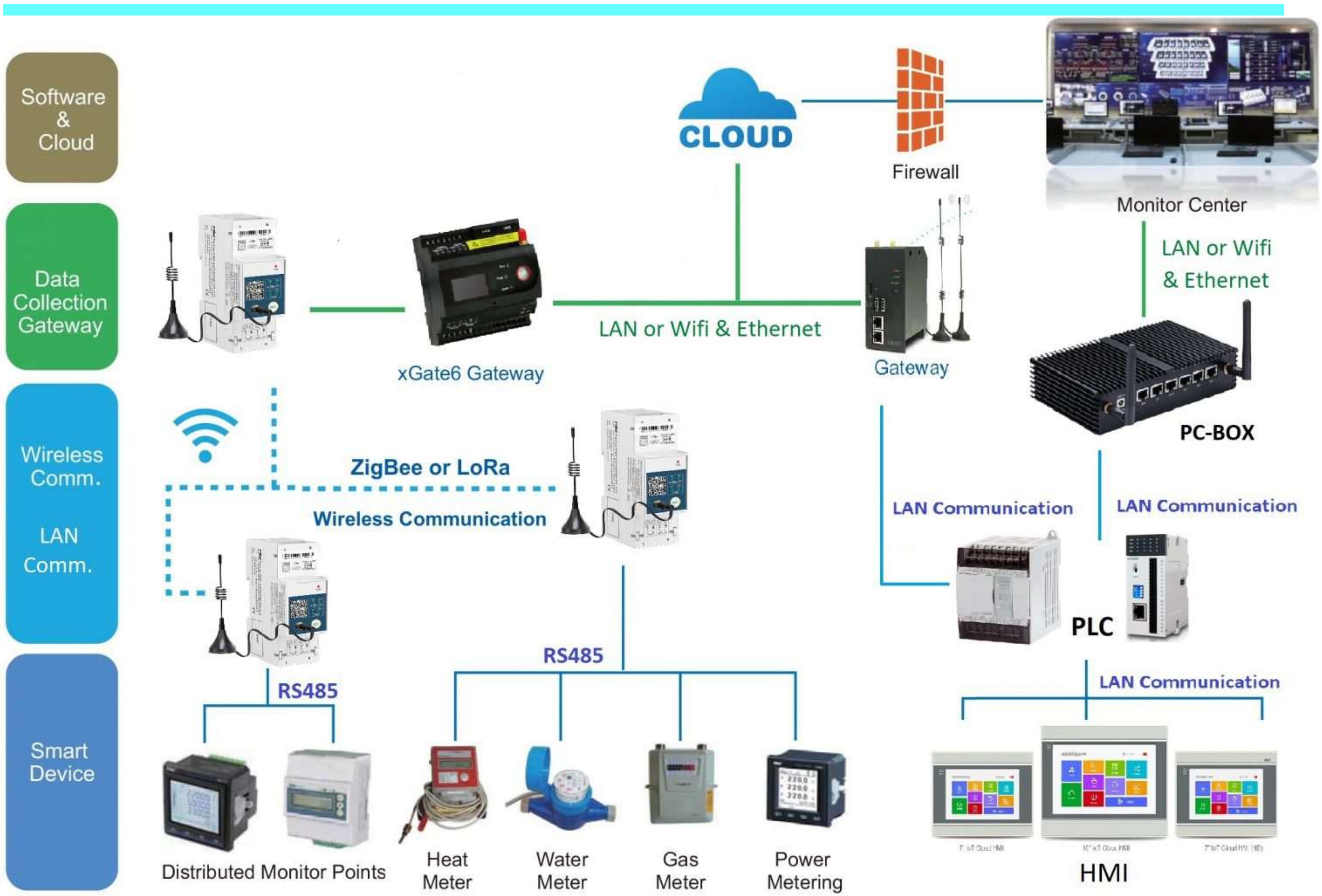
http://www.hi-flying.com/

Elfin Series Assembly Drawing

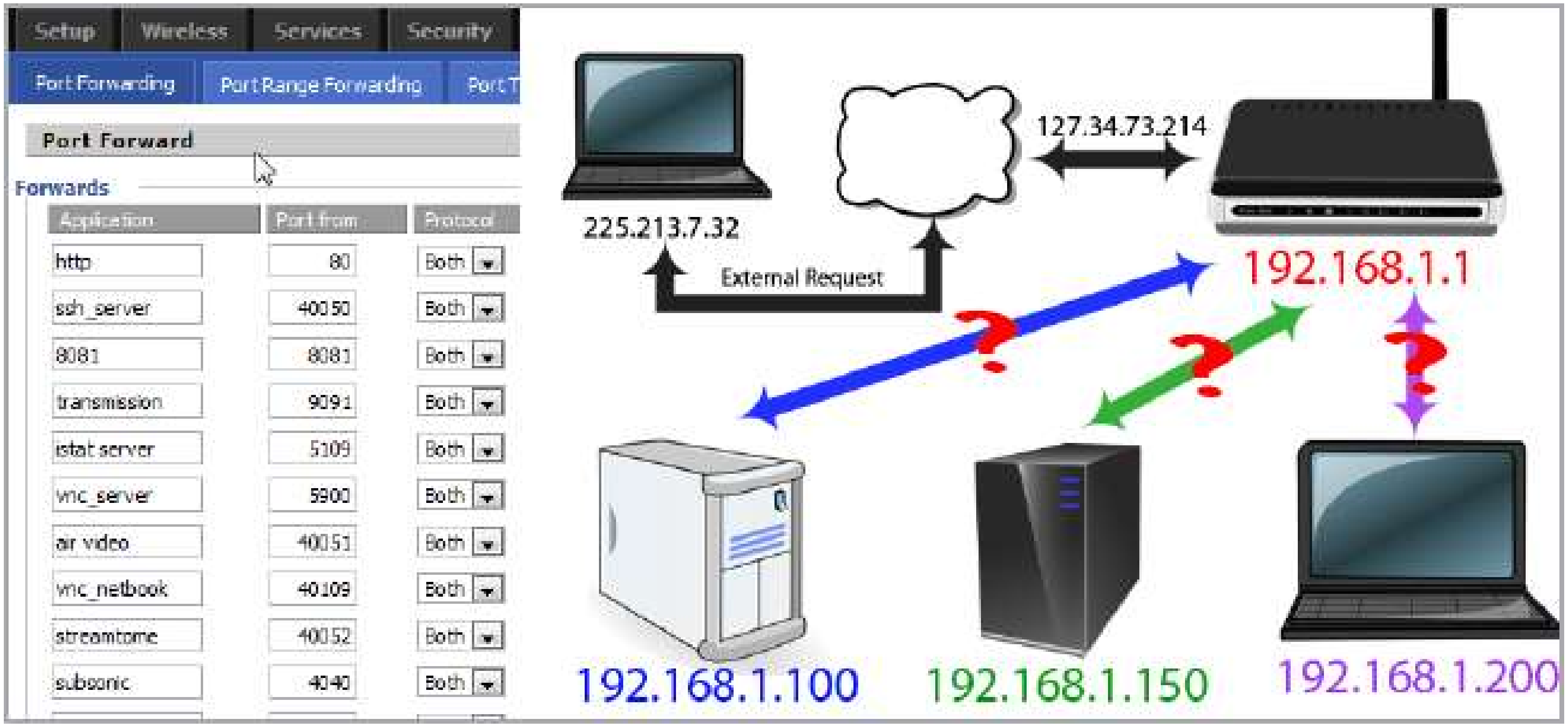
https://www.youtube.com/watch?v=iSv0HaSmPRo



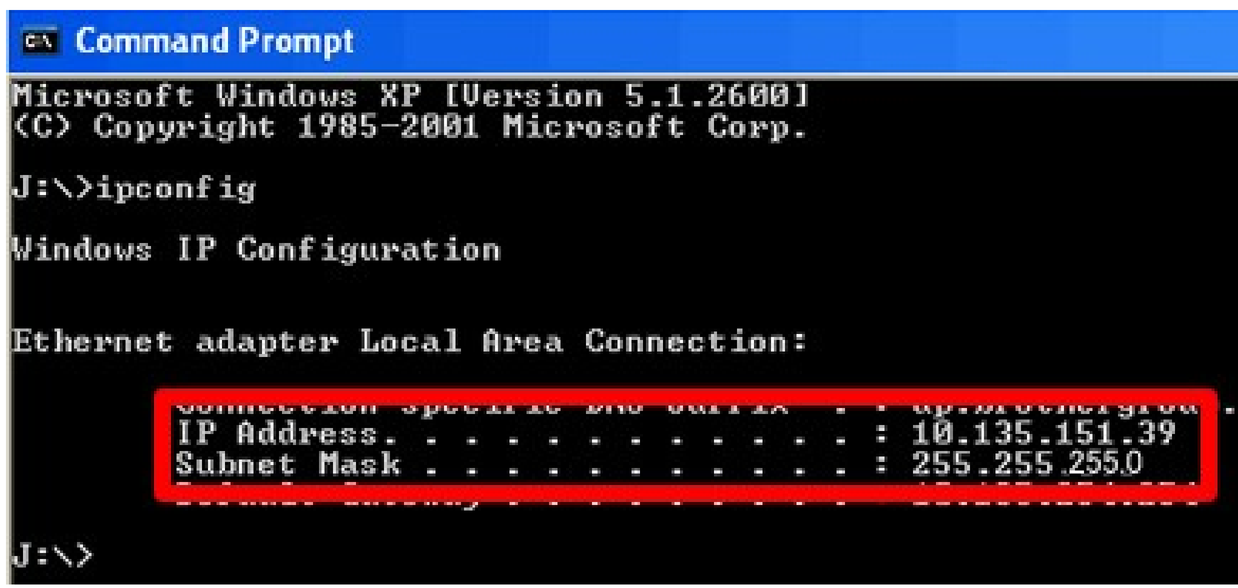

Web Browser - Remote
Monitor & Control
SCADA
Supervisory Controls
SCADA
Supervisory Controls
TCP/IP
Modbus TCP
Client
Ethernet
Modbus TCP
Server 255
Modbus TCP
Server 3
VFD
Modbus TCP
Server 2
Operator
Interface
Modbus RTU/ASCII
RS485
Operator
Interface
Temperature
Controller
Modbus

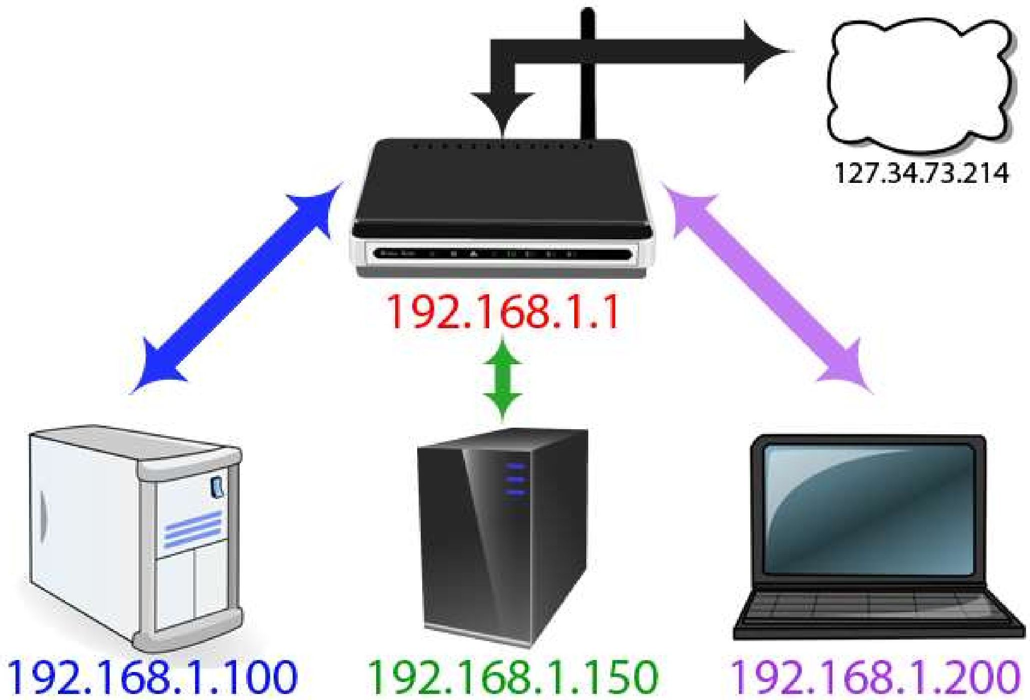


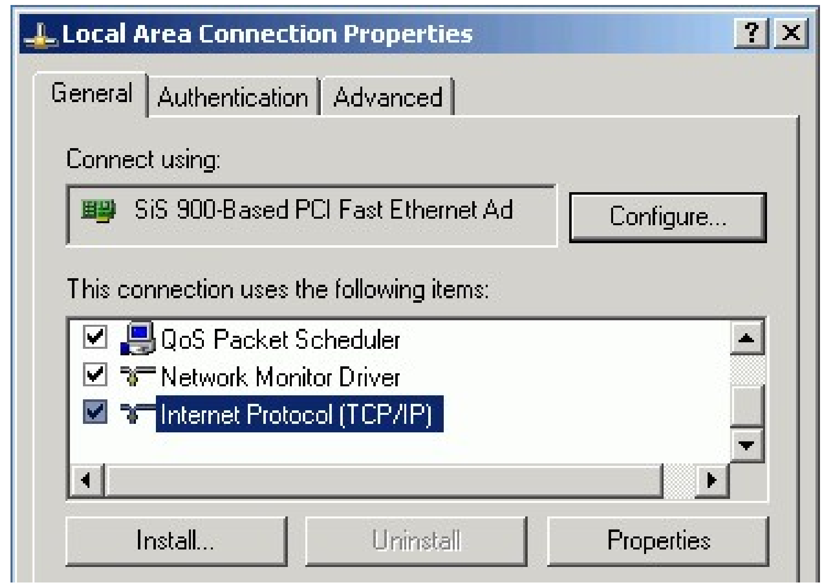
Software & Cloud
Data Collection Gateway
Wireless Comm.
LAN Comm.
Smart Device
CLOUD
Firewall
Monitor Center
LAN or Wifi & Ethernet
xGate6 Gateway
LAN or Wifi & Ethernet
Gateway
PC-BOX
ZigBee or LoRa
Wireless Communication
LAN Communication
LAN Communication
PLC
RS485
RS485
LAN Communication
Distributed Monitor Points
Heat Meter
Water Meter
Gas Meter
Power Metering
HMI


Setup
Wireless
Services
Security
Port Forwarding
Port Range Forwarding
Port Forward
Forwards
Application
Port from
Protocol
http
80
Both
ssh_server
40050
Both
8081
8081
Both
transmission
9091
Both
istat server
5109
Both
vnc_server
5900
Both
air video
40051
Both
vnc_netbook
40109
Both
streamtome
40052
Both
subsonic
4040
Both
225.213.7.32
127.34.73.214
External Request
192.168.1.1
192.168.1.100
192.168.1.150
192.168.1.200

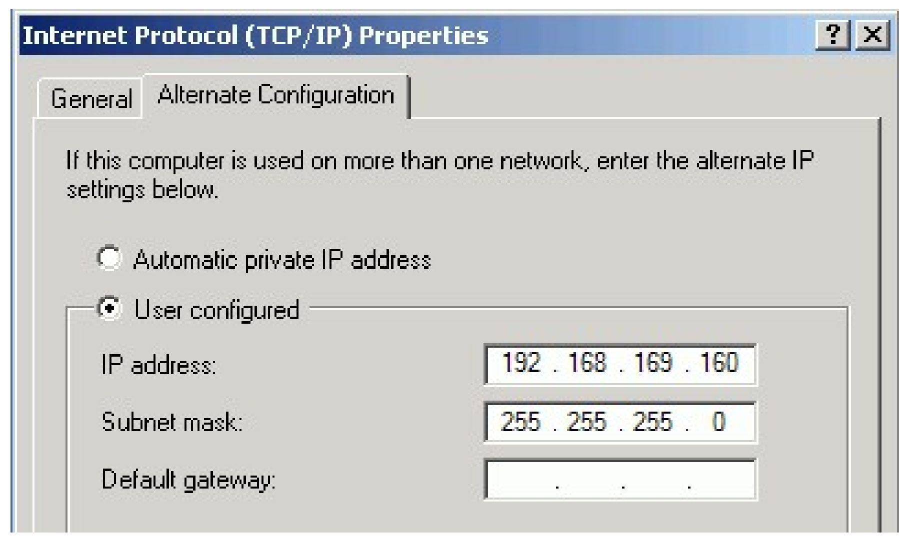
ipconfig


Command Prompt
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.
J:\>ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection:
IP Address. . . . . : 10.135.151.39
Subnet Mask . . . . : 255.255.255.0
J:\>

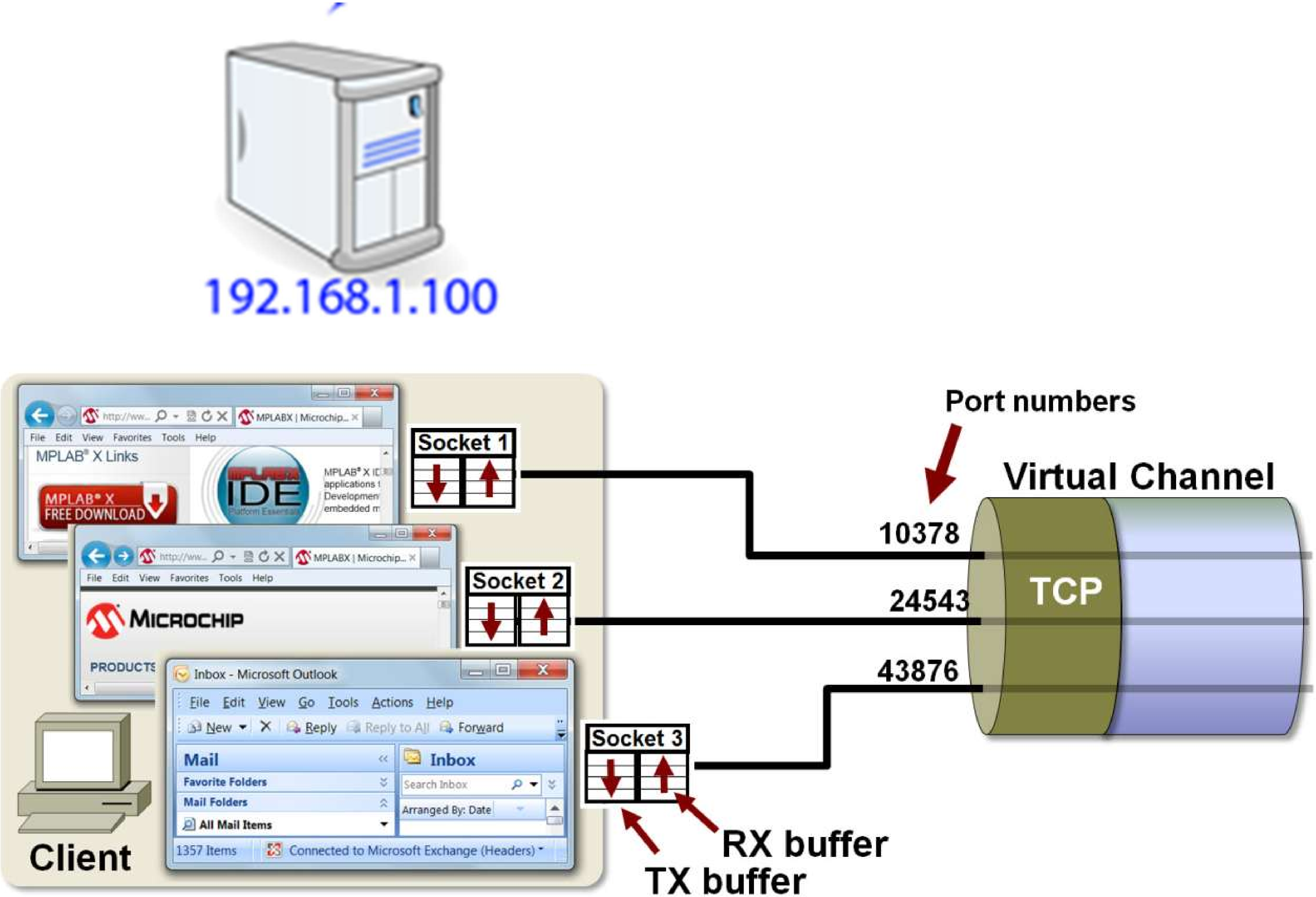





```
C:\Documents and Settings\demo>ipconfig

Windows IP Configuration


Ethernet adapter Local Area Connection:

        Connection-specific DNS Suffix  . : racom.cz
        IP Address. . . . . . . . . . . . : 192.168.169.160
        Subnet Mask . . . . . . . . . . . : 255.255.255.0
        Default Gateway . . . . . . . . . :
```




```
C:\Documents and Settings\demo>ping 192.168.169.169

Pinging 192.168.169.169 with 32 bytes of data:

Reply from 192.168.169.169: bytes=32 time<1ms TTL=64
Reply from 192.168.169.169: bytes=32 time<1ms TTL=64
Reply from 192.168.169.169: bytes=32 time<1ms TTL=64
Reply from 192.168.169.169: bytes=32 time<1ms TTL=64
```


192.168.1.100
Port numbers
Virtual Channel
Socket 1
Socket 2
Socket 3
10378
24543
43876
TCP
RX buffer
TX buffer
Client

การใช้งานโปรแกรม **SimModbus 2** จำลอง **Modbus Memory Device (TCP)**

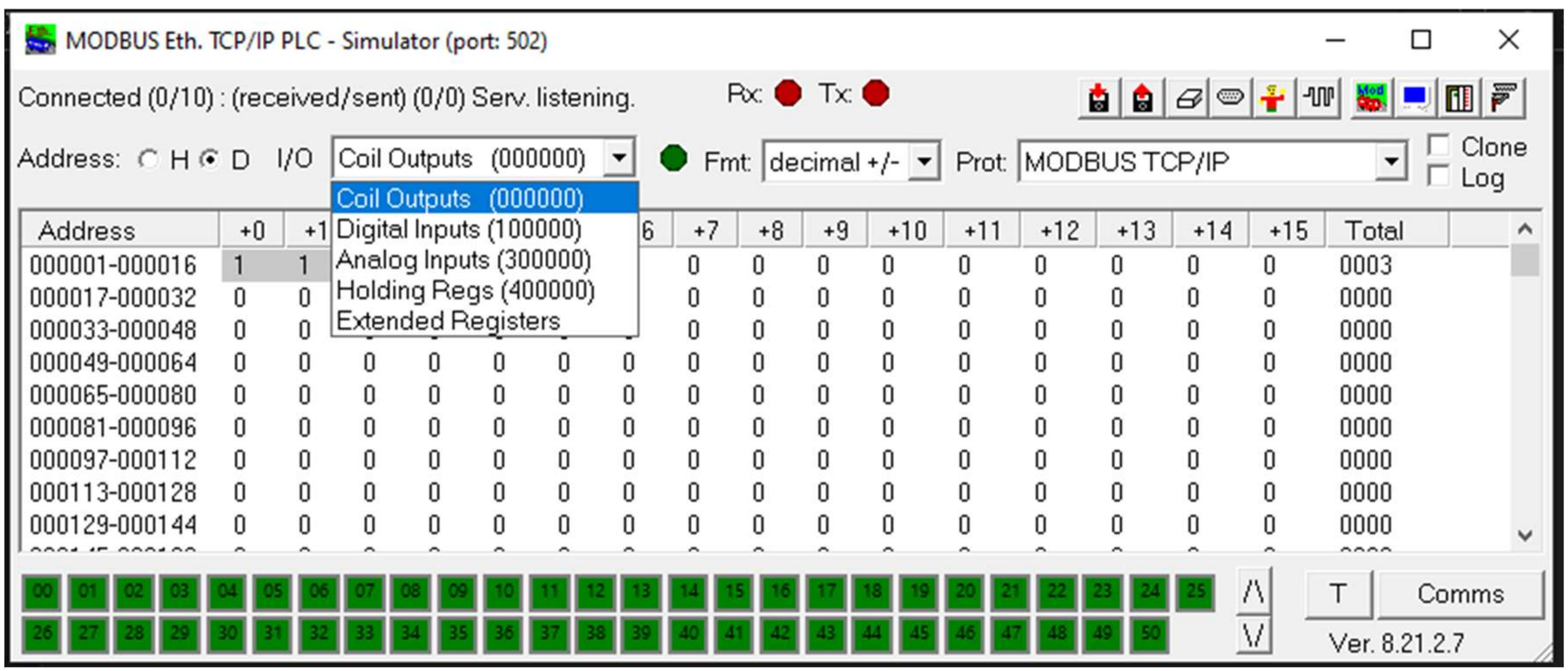
ตำแหน่งหน่วยความจำ ภายในอุปกรณ์

000001 - 065535 Coil Output	100000 -> 165535 Digital Input
300000 –365535 Analog Input	400000 – 465535 Holding Reg

Read / Write **Read Only**

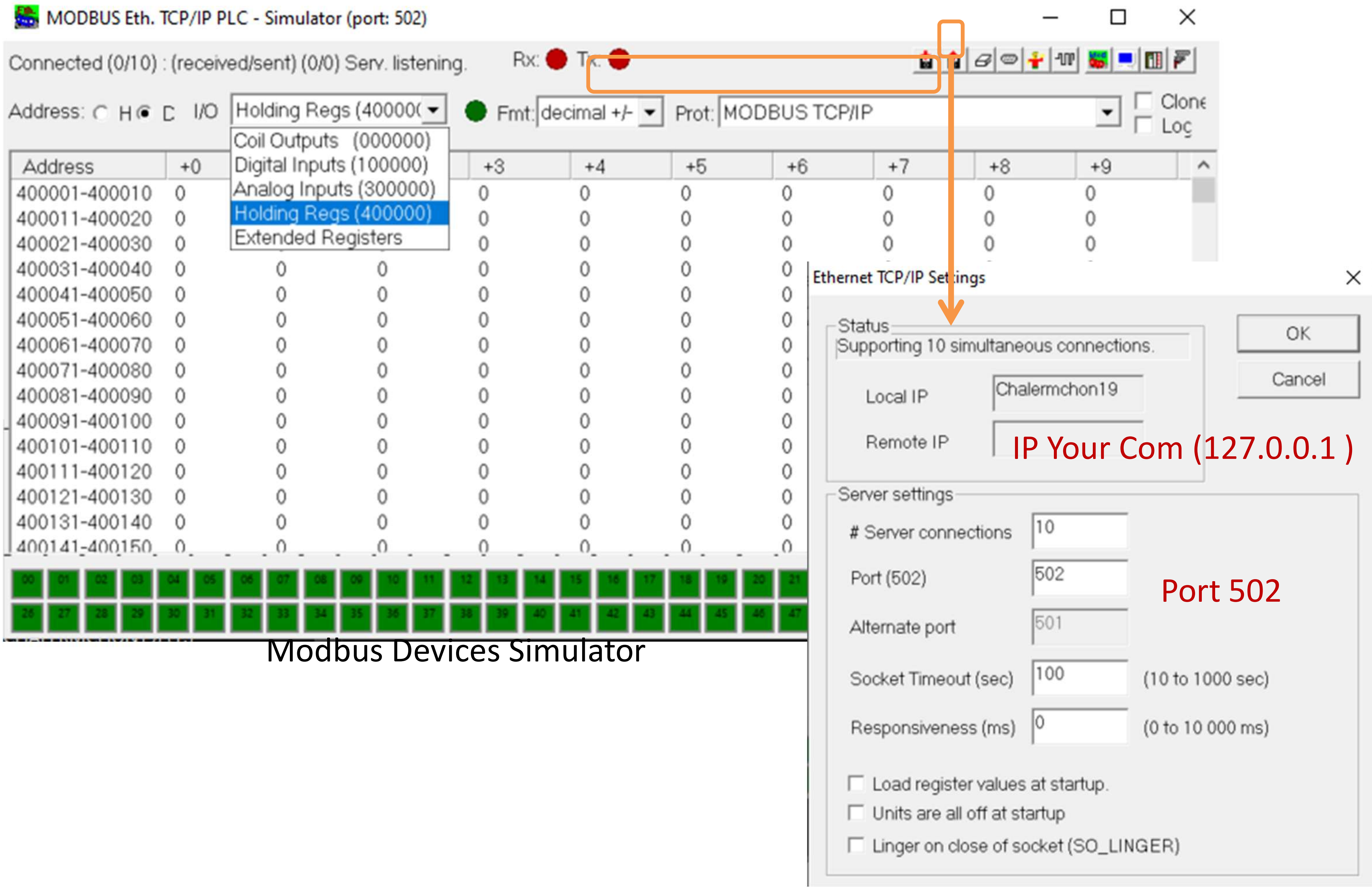
เก็บค่าตัวเลข
0 และ 1

เก็บค่าจำนวนเต็ม
(0- FFFFH)
(0- 65535)

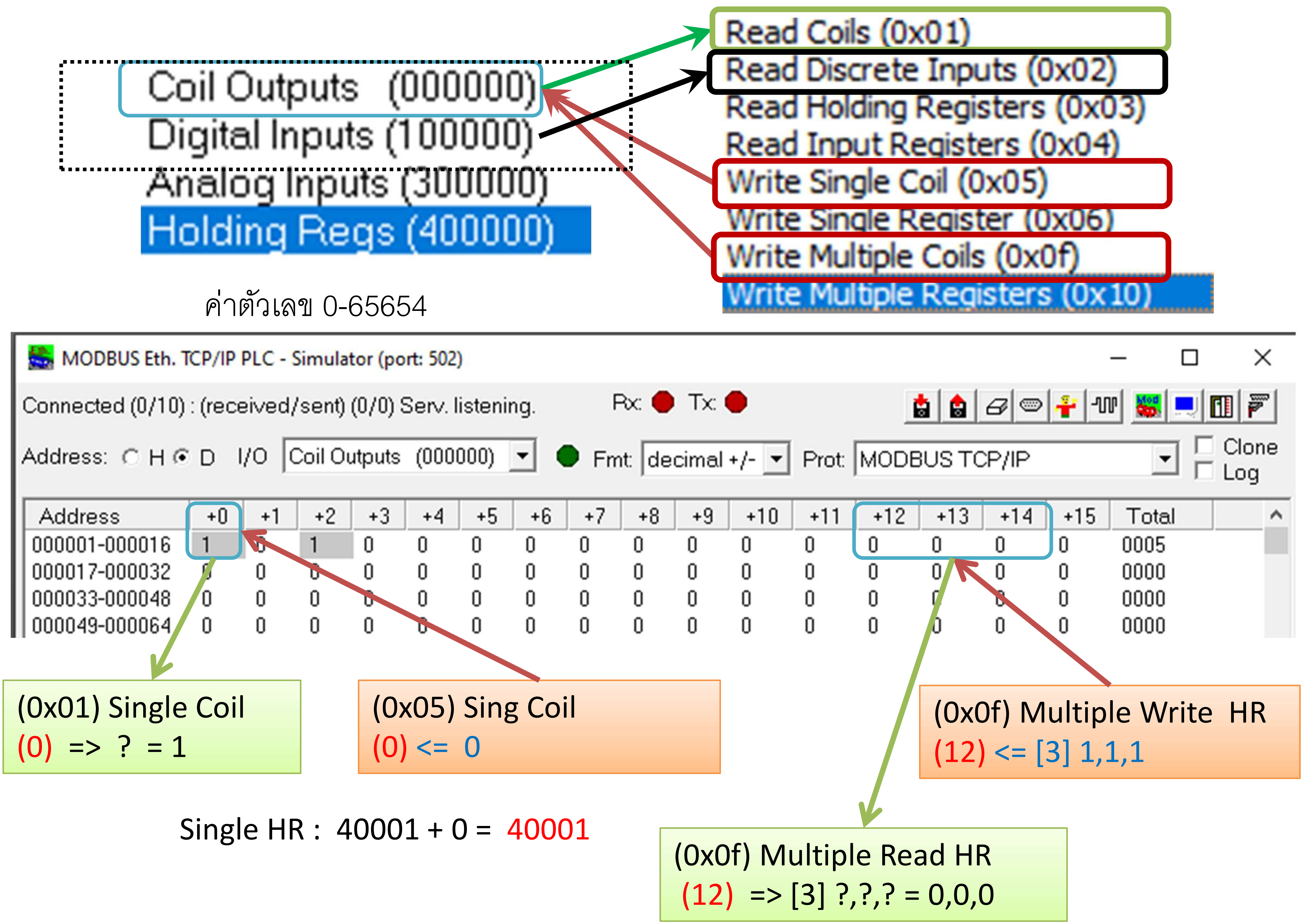


YouTube Modbus - 01 SimModbus

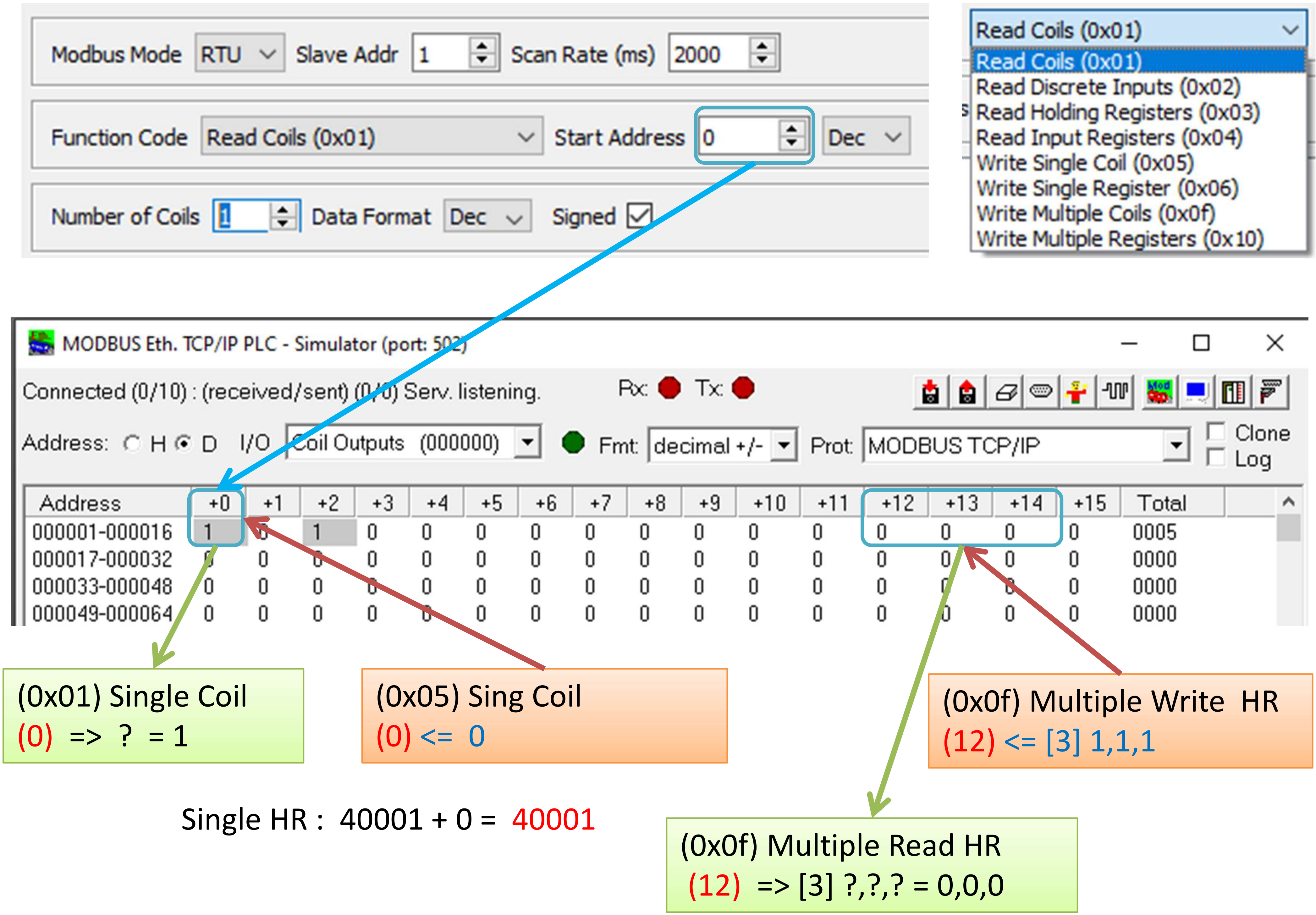
การใช้งานโปรแกรม SimModbus 2

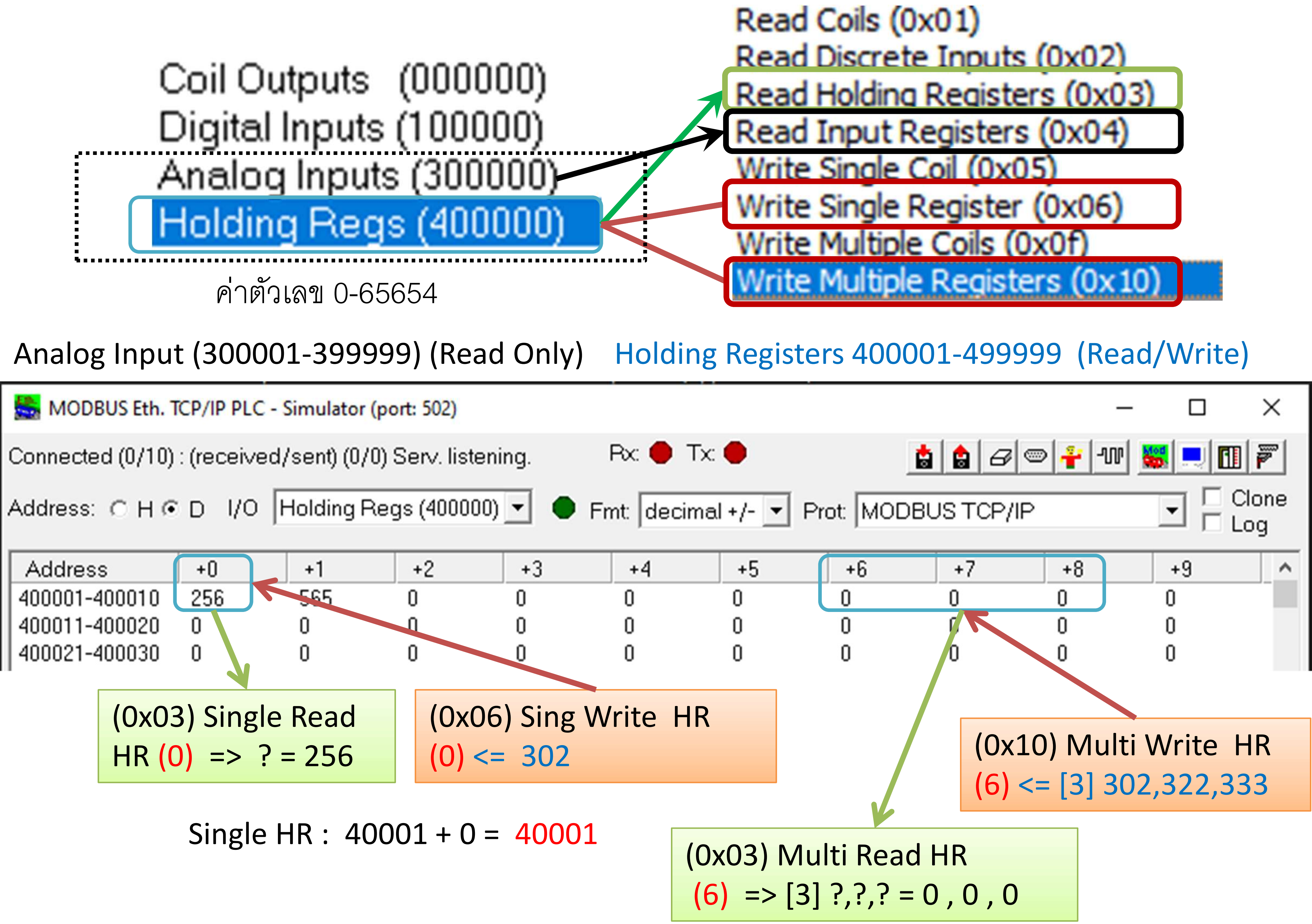


Modbus Devices Simulator



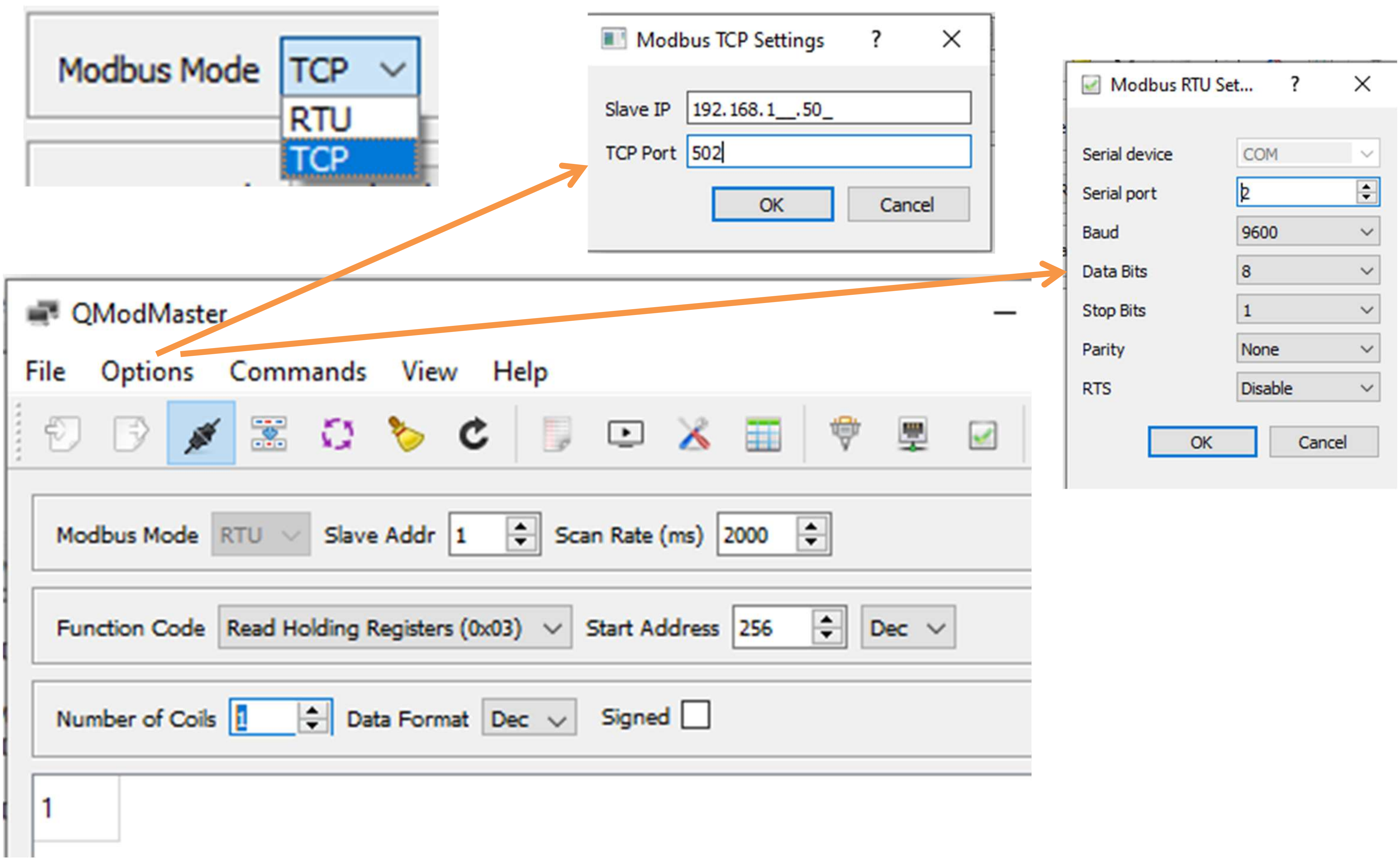
Coil Outputs (000000)
Digital Inputs (100000)
Analog Inputs (300000)
Holding Regs (400000)
Read Coils (0x01)
Read Discrete Inputs (0x02)
Read Holding Registers (0x03)
Read Input Registers (0x04)
Write Single Coil (0x05)
Write Single Register (0x06)
Write Multiple Coils (0x0f)
Write Multiple Registers (0x10)
ค่าตัวเลข 0-65654
MODBUS Eth. TCP/IP PLC - Simulator (port: 502)
Connected (0/10) : (received/sent) (0/0) Serv. listening.
Rx: Tx:
Address: H D I/O Coil Outputs (000000) Fmt: decimal +/- Prot: MODBUS TCP/IP Clone Log
Address +0 +1 +2 +3 +4 +5 +6 +7 +8 +9 +10 +11 +12 +13 +14 +15 Total
000001-000016 1 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0005
000017-000032 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0000
000033-000048 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0000
000049-000064 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0000
(0x01) Single Coil
(0) => ? = 1
(0x05) Sing Coil
(0) <= 0
(0x0f) Multiple Write HR
(12) <= [3] 1,1,1
Single HR : 40001 + 0 = 40001
(0x0f) Multiple Read HR
(12) => [3] ?,?,? = 0,0,0

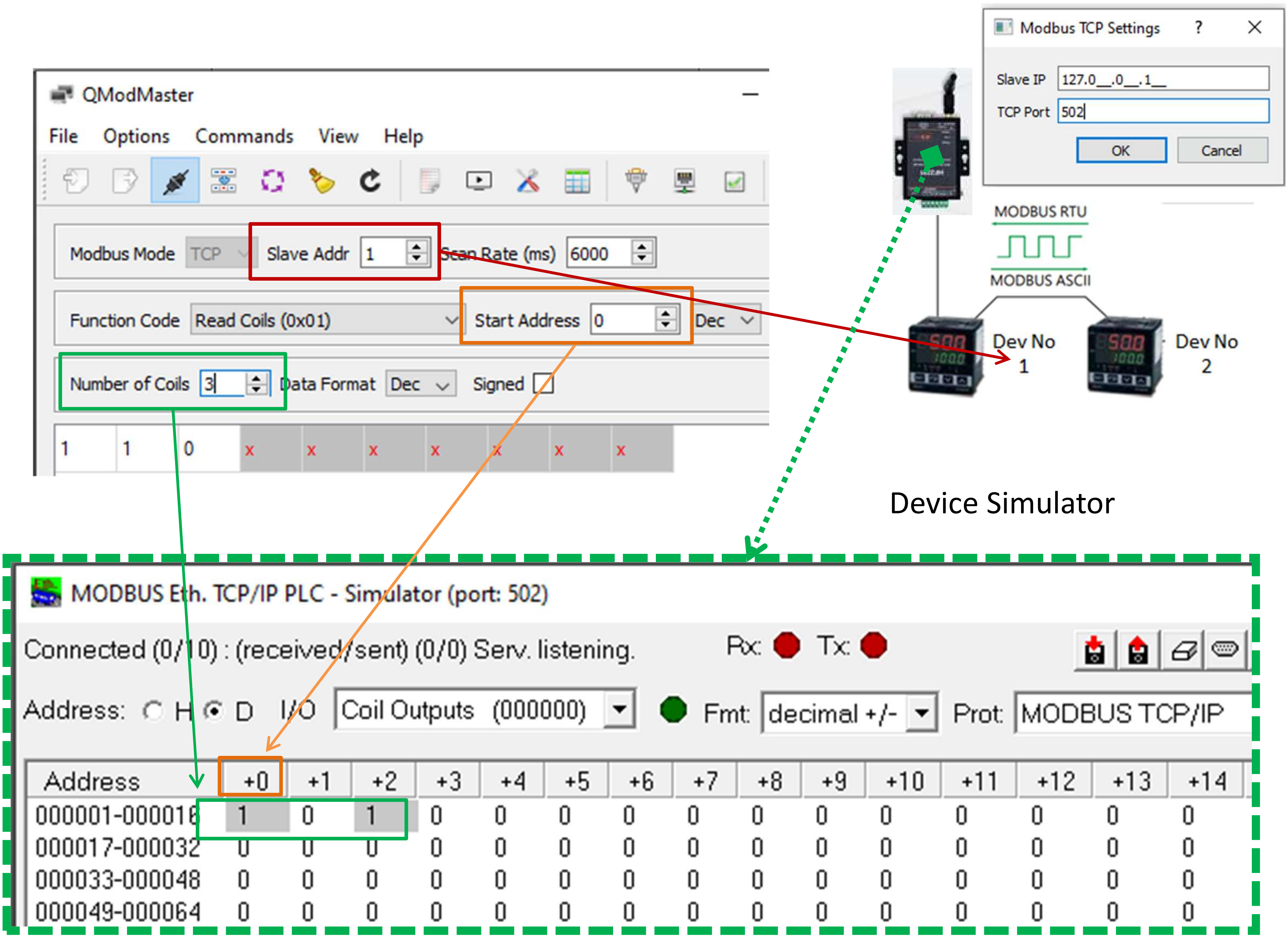

Modbus Mode RTU
Slave Addr 1
Scan Rate (ms) 2000
Function Code Read Coils (0x01)
Start Address 0
Dec
Number of Coils 1
Data Format Dec
Signed
Read Coils (0x01)
Read Coils (0x01)
Read Discrete Inputs (0x02)
Read Holding Registers (0x03)
Read Input Registers (0x04)
Write Single Coil (0x05)
Write Single Register (0x06)
Write Multiple Coils (0x0f)
Write Multiple Registers (0x10)
MODBUS Eth. TCP/IP PLC - Simulator (port: 502)
Connected (0/10) : (received/sent) (0/0) Serv. listening.
Rx: Tx:
Address: H D I/O Coil Outputs (000000)
Fmt: decimal +/-
Prot: MODBUS TCP/IP
Clone
Log
Address +0 +1 +2 +3 +4 +5 +6 +7 +8 +9 +10 +11 +12 +13 +14 +15 Total
000001-000016 1 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0005
000017-000032 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0000
000033-000048 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0000
000049-000064 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0000
(0x01) Single Coil
(0) => ? = 1
(0x05) Sing Coil
(0) <= 0
(0x0f) Multiple Write HR
(12) <= [3] 1,1,1
Single HR : 40001 + 0 = 40001
(0x0f) Multiple Read HR
(12) => [3] ?,?,? = 0,0,0



Coil Outputs (000000)
Digital Inputs (100000)
Analog Inputs (300000)
Holding Regs (400000)
ค่าตัวเลข 0-65654
Read Coils (0x01)
Read Discrete Inputs (0x02)
Read Holding Registers (0x03)
Read Input Registers (0x04)
Write Single Coil (0x05)
Write Single Register (0x06)
Write Multiple Coils (0x0f)
Write Multiple Registers (0x10)
Analog Input (300001-399999) (Read Only)
Holding Registers 400001-499999 (Read/Write)
MODBUS Eth. TCP/IP PLC - Simulator (port: 502)
Connected (0/10) : (received/sent) (0/0) Serv. listening.
Rx: Tx:
Address: H D I/O Holding Regs (400000) Fmt: decimal +/- Prot: MODBUS TCP/IP
Clone
Log
Address +0 +1 +2 +3 +4 +5 +6 +7 +8 +9
400001-400010 256 565 0 0 0 0 0 0 0 0
400011-400020 0 0 0 0 0 0 0 0 0 0
400021-400030 0 0 0 0 0 0 0 0 0 0
(0x03) Single Read HR (0) => ? = 256
(0x06) Sing Write HR (0) <= 302
(0x10) Multi Write HR (6) <= [3] 302,322,333
Single HR : 40001 + 0 = 40001
(0x03) Multi Read HR (6) => [3] ?,?,? = 0 , 0 , 0

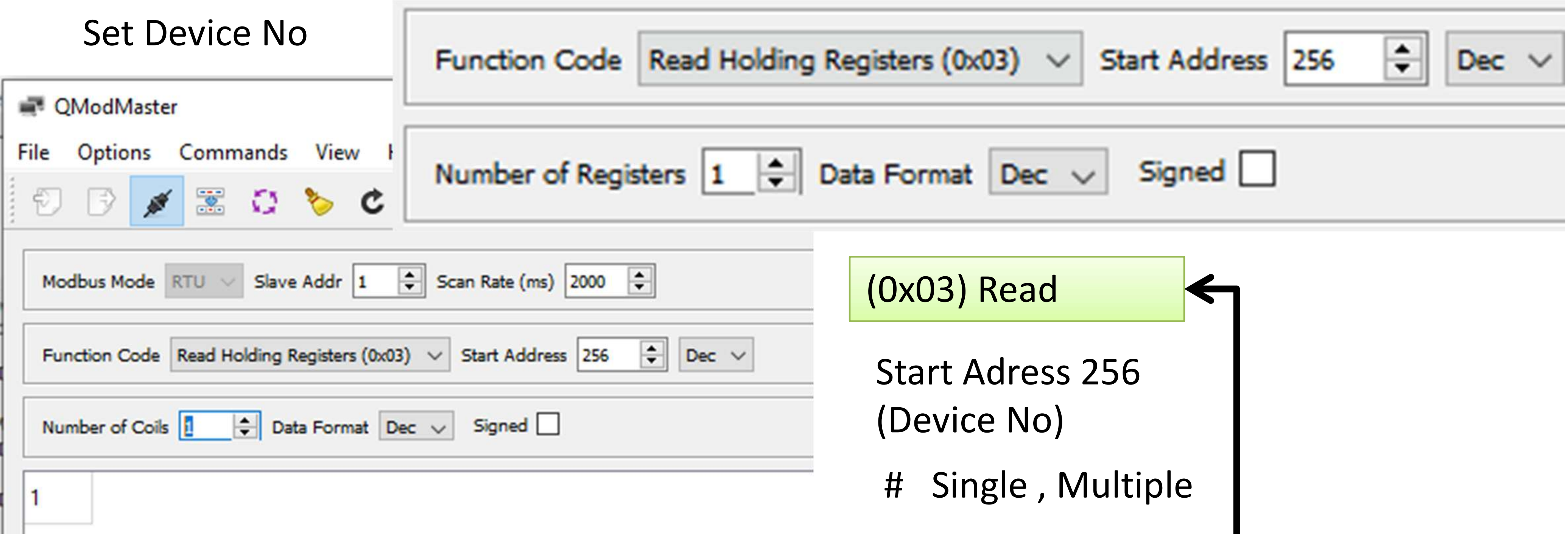
การใช้งานโปรแกรม QModMaster (Modbus Master)




QModMaster
File Options Commands View Help
Modbus Mode TCP
Slave Addr 1
Scan Rate (ms) 6000
Function Code Read Coils (0x01)
Start Address 0
Dec
Number of Coils 3
Data Format Dec
Signed
1 1 0 x x x x x x x
Modbus TCP Settings
Slave IP 127.0__.0__.1__
TCP Port 502
OK
Cancel
MODBUS RTU
MODBUS ASCII
Dev No 1
Dev No 2
Device Simulator
MODBUS Eth. TCP/IP PLC - Simulator (port: 502)
Connected (0/10) : (received/sent) (0/0) Serv. listening.
Rx: Tx:
Address: H D I/O Coil Outputs (000000)
Fmt: decimal +/-
Prot: MODBUS TCP/IP
Address +0 +1 +2 +3 +4 +5 +6 +7 +8 +9 +10 +11 +12 +13 +14
000001-000016 1 0 1 0 0 0 0 0 0 0 0 0 0 0 0
000017-000032 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
000033-000048 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
000049-000064 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

YouTube Modbus – 02 QModMaster

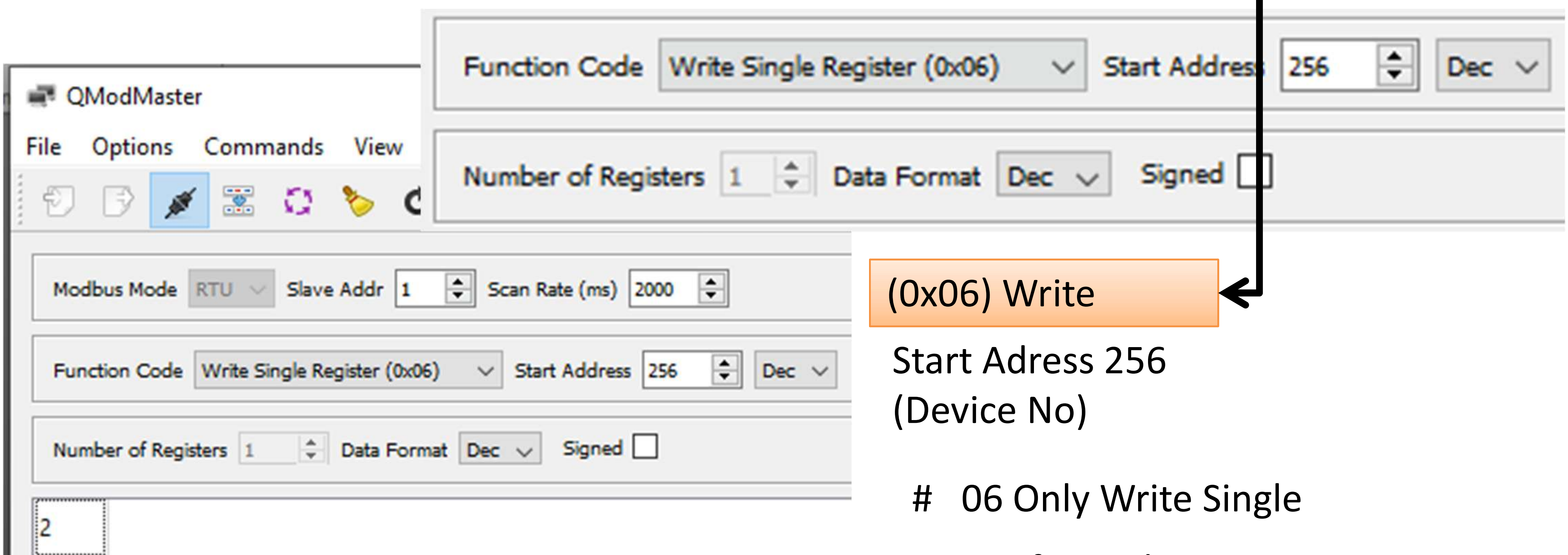
Set Device No



(0x03) Read

Start Adress 256
(Device No)

Single , Multiple



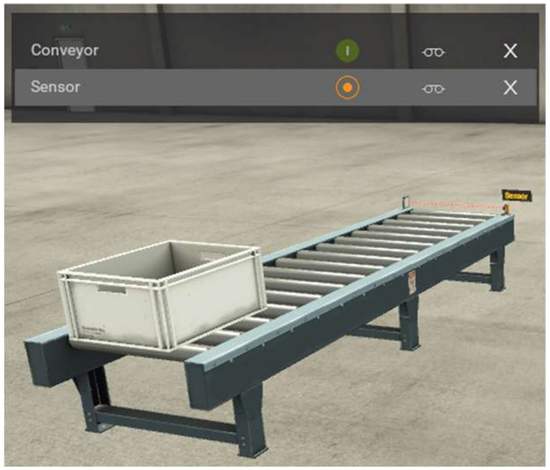
(0x06) Write

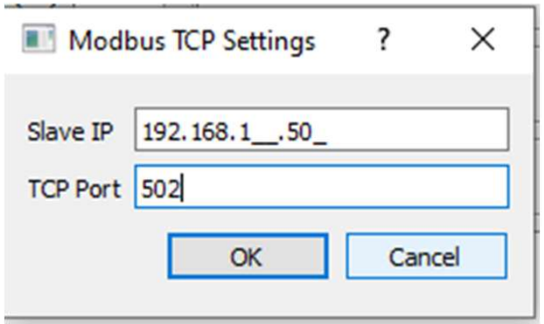
Start Adress 256
(Device No)

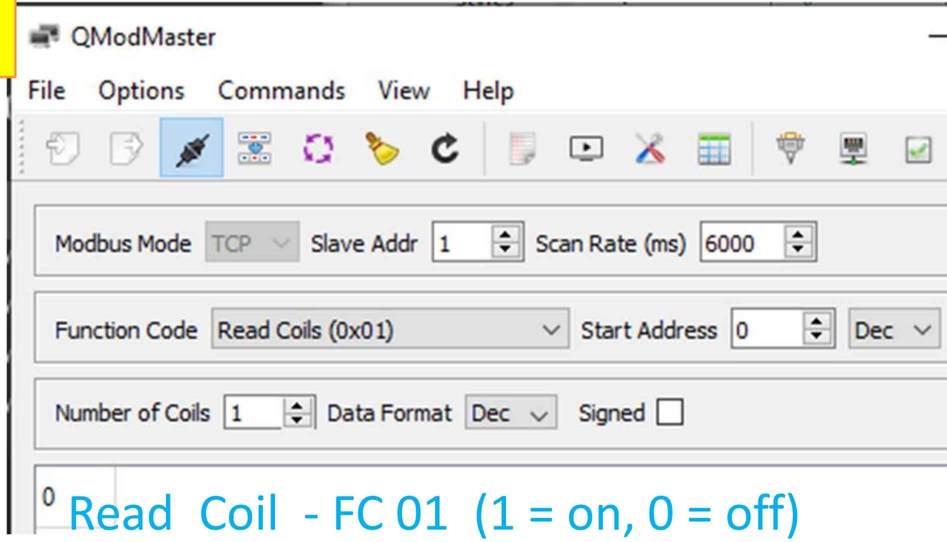
06 Only Write Single

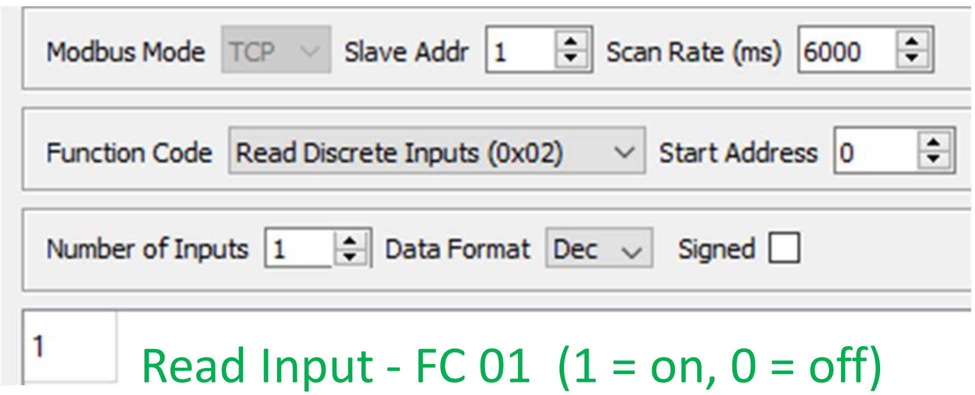
60 for Multi Write

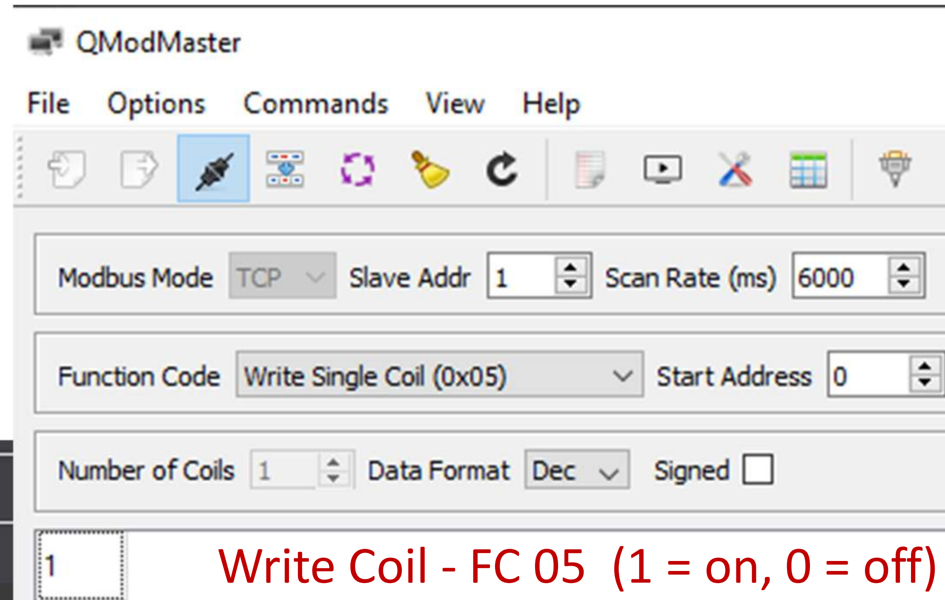
YouTube Modbus – 03 FactoryIO Conveyor

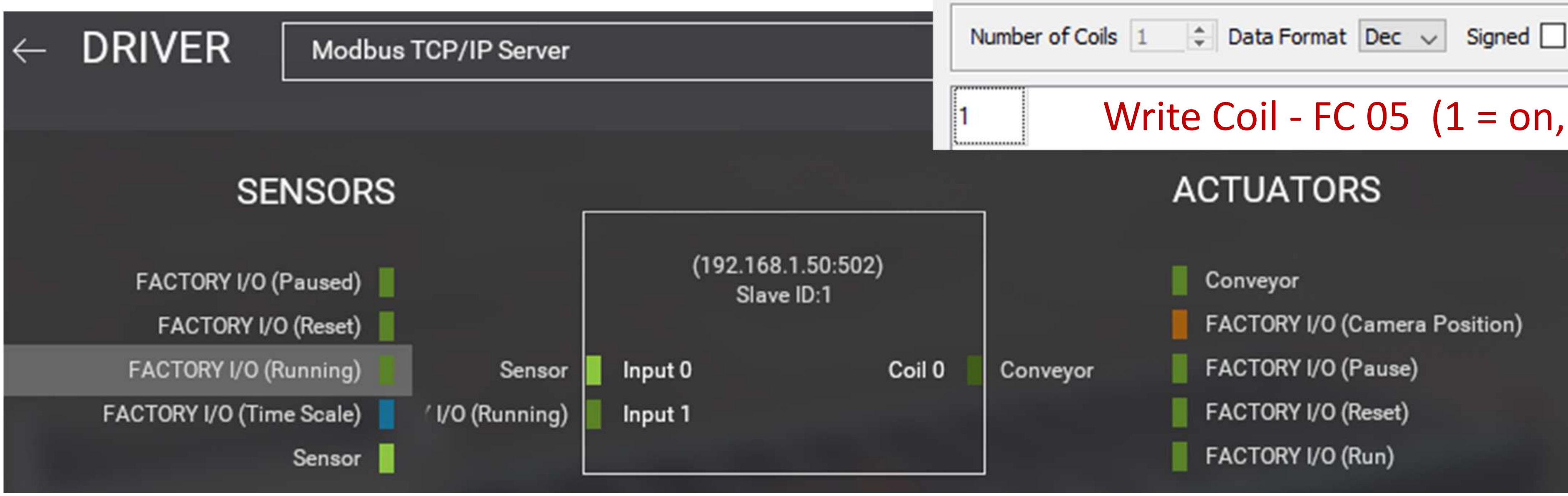




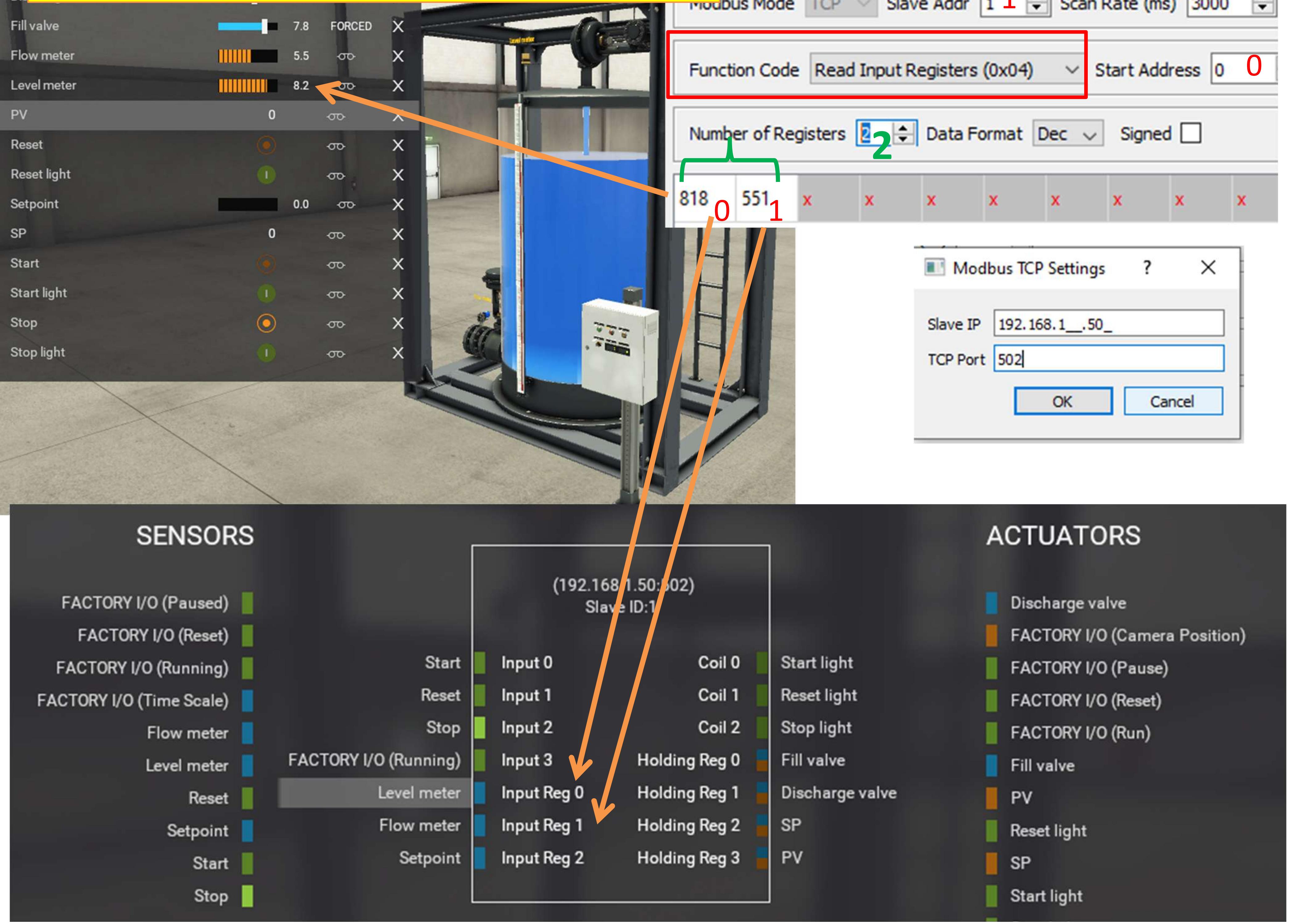




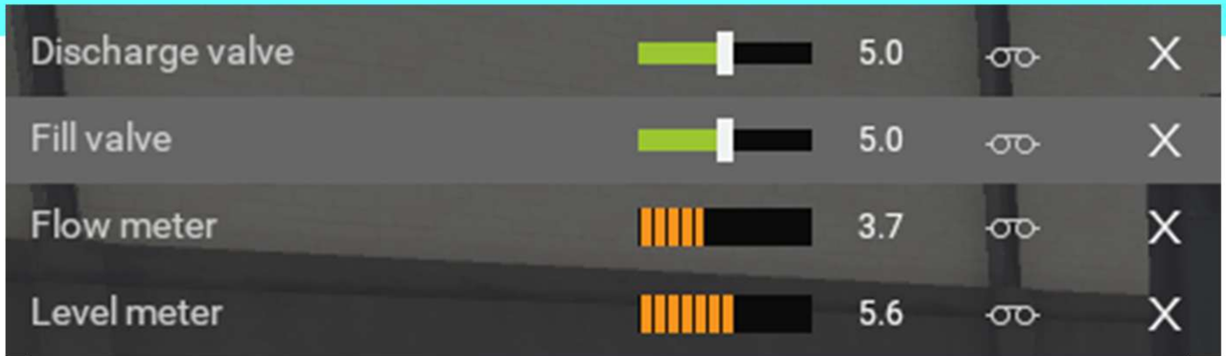




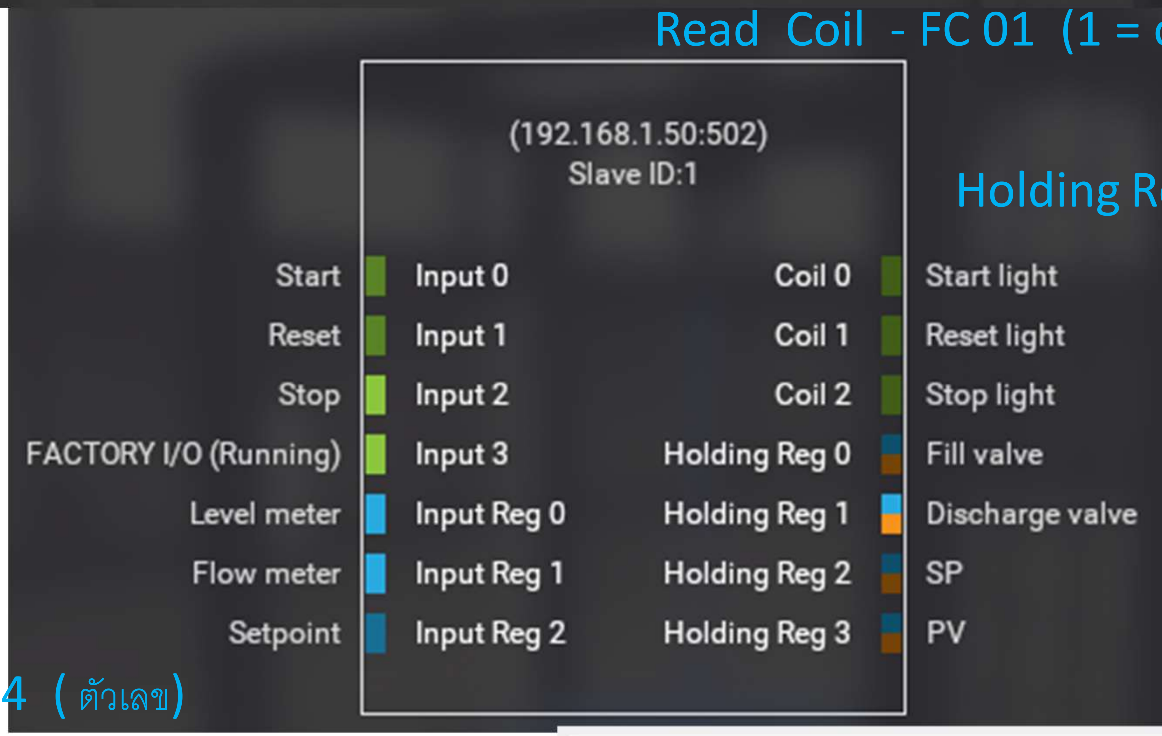
YouTube Modbus – 04 Level Control



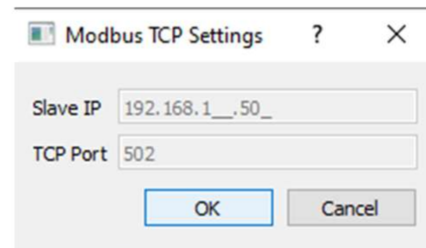
การติดต่อ Modbus Machine with QModMaster



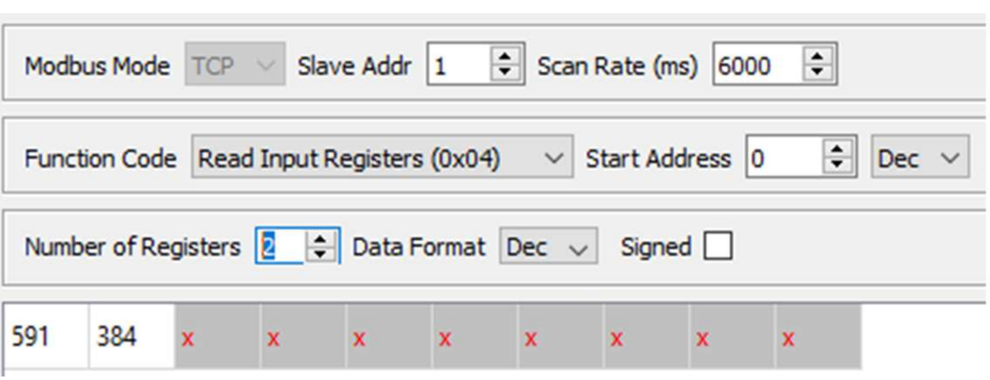
Read Coil - FC 01 (1 = on, 0 = off)

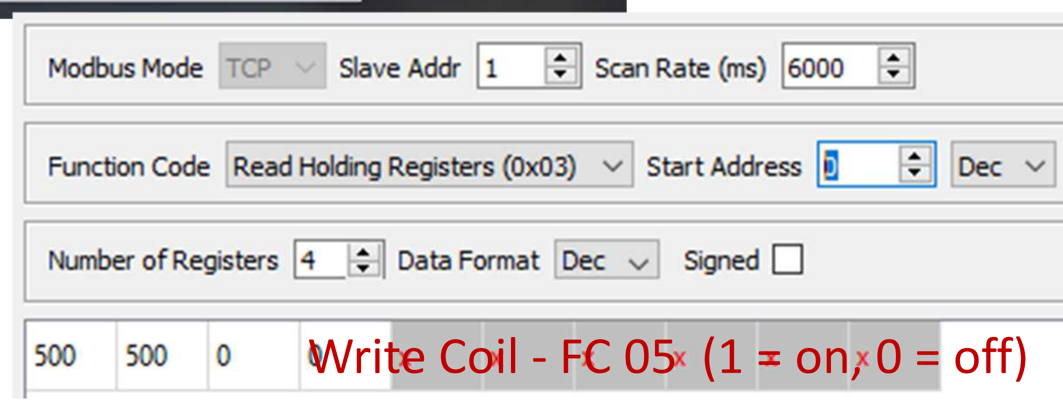


Holding Reg - FC 03 (ตัวเลข)



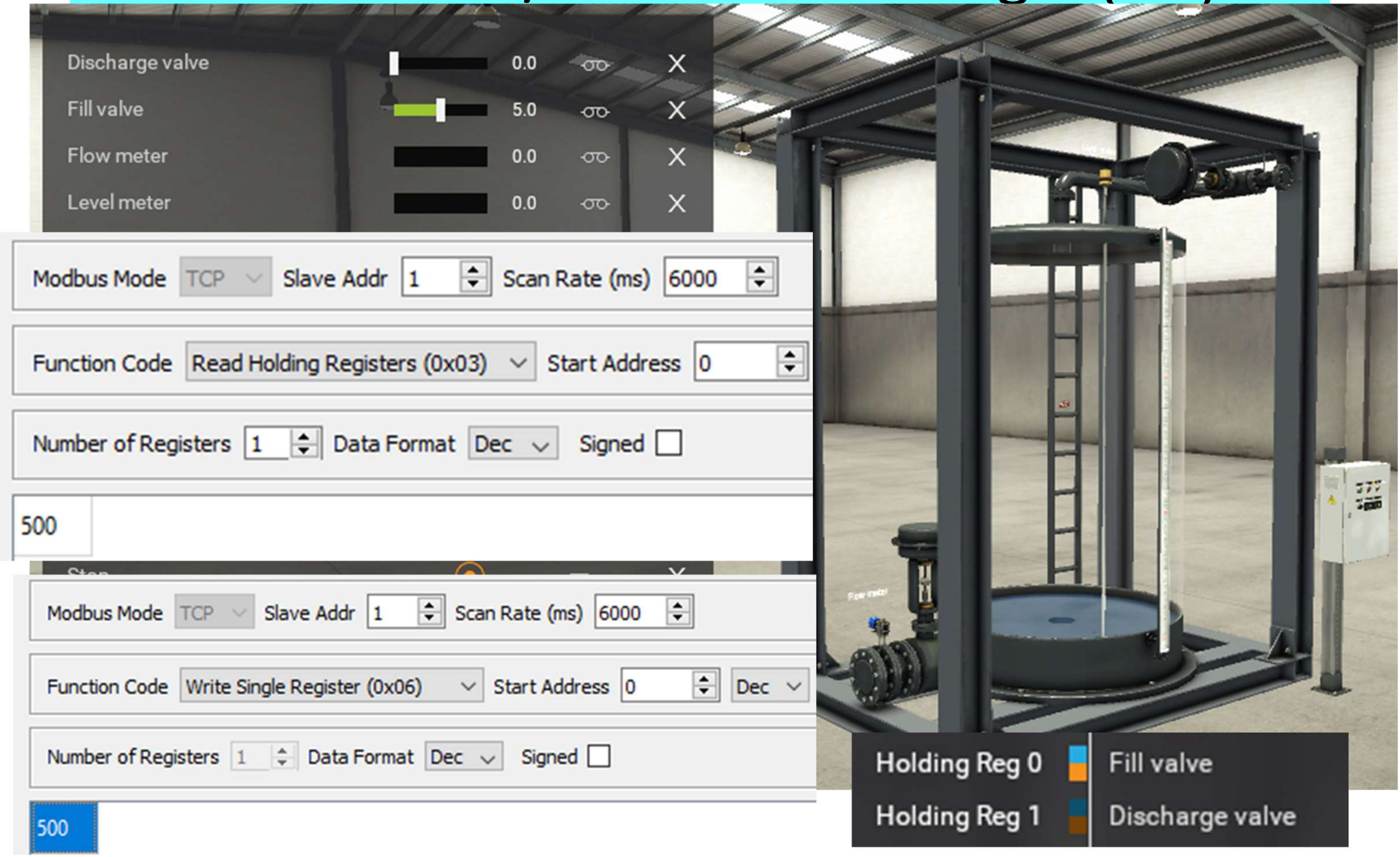
Read Input Reg - FC 04 (ตัวเลข)

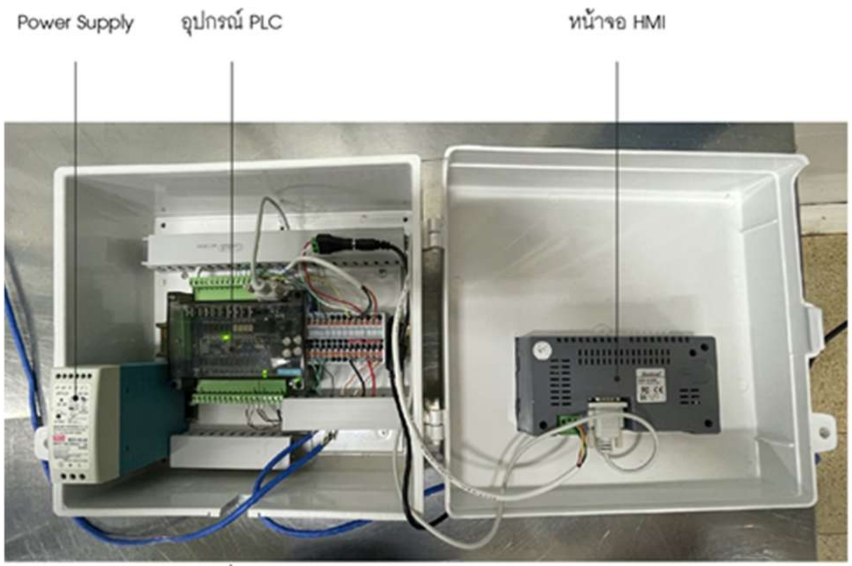




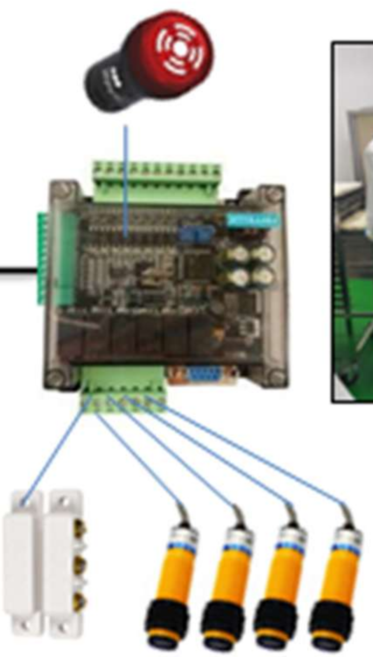
Write Coil - FC 05 (1 = on, 0 = off)

FC 03 Read / FC 06 Write Single (HR)



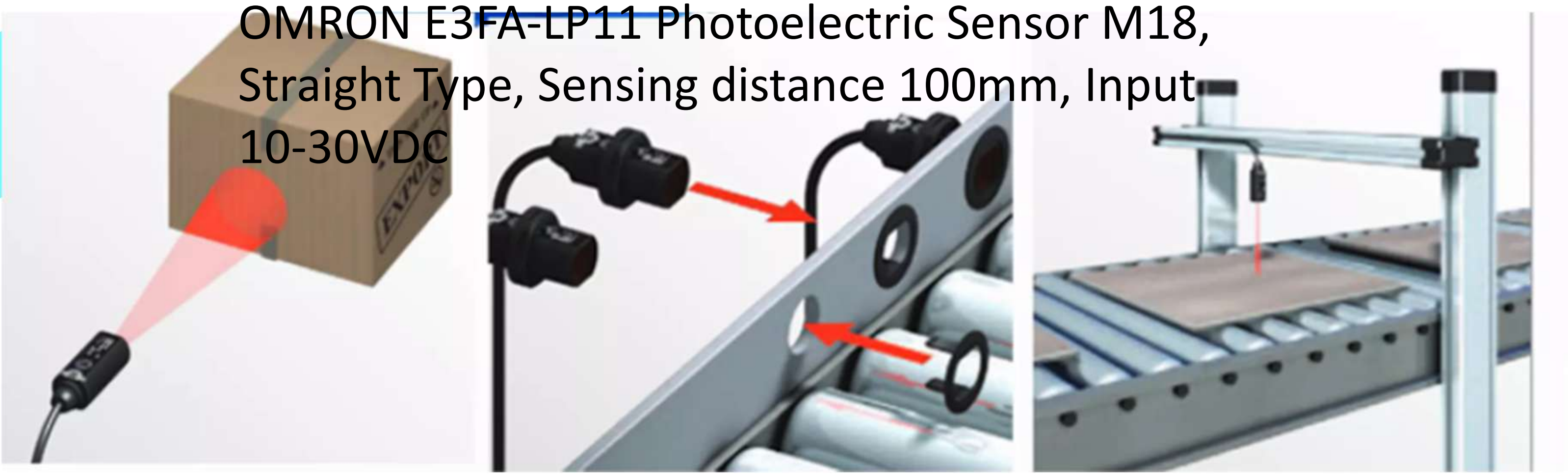

Power Supply
อุปกรณ์ PLC
หน้าจอ HMI

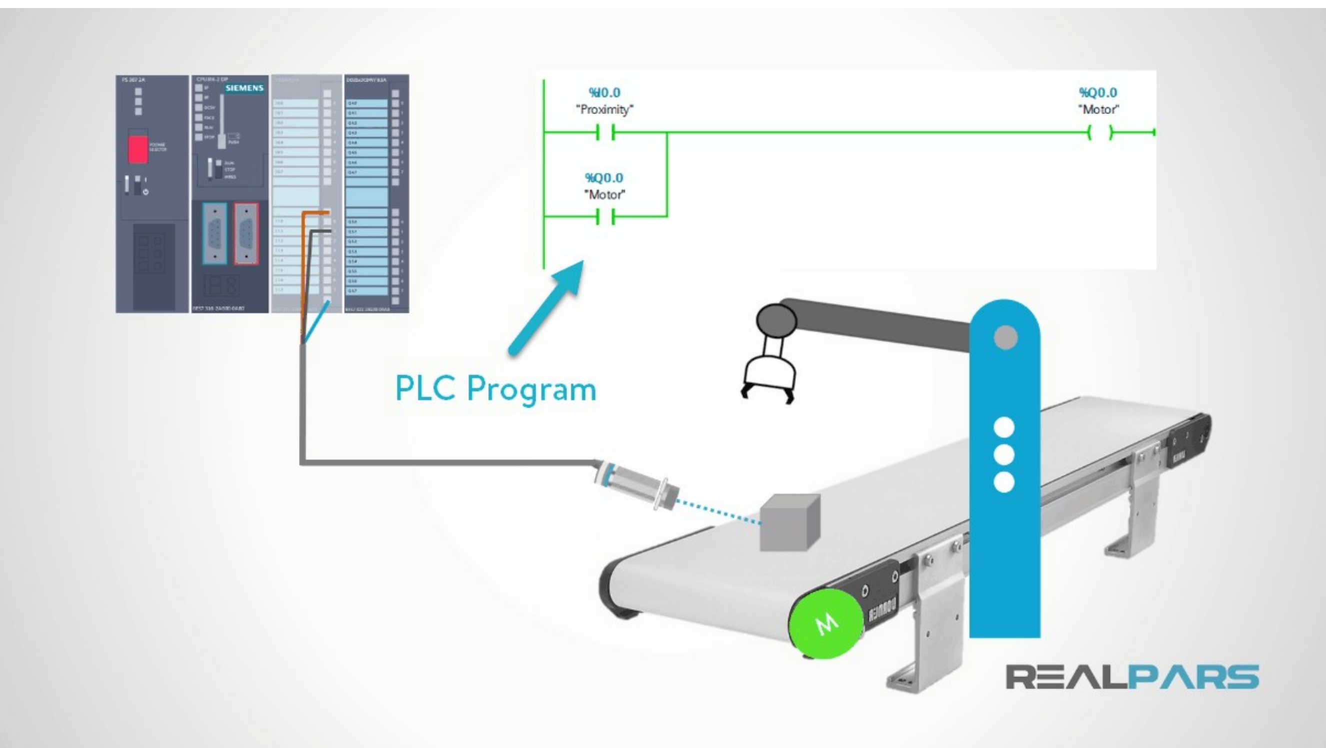

ปุ่ม Reset
จอแสดงจำนวน
การย่างสาหร่าย
แต่ละเตา
ปุ่มปรับเวลา Delay
ปุ่มปรับลดจำนวน
ปุ่มปรับระดับเสียง
1
100
65525
Setting
111
Sensor Delay
2
2
88
23
Sound Level
5



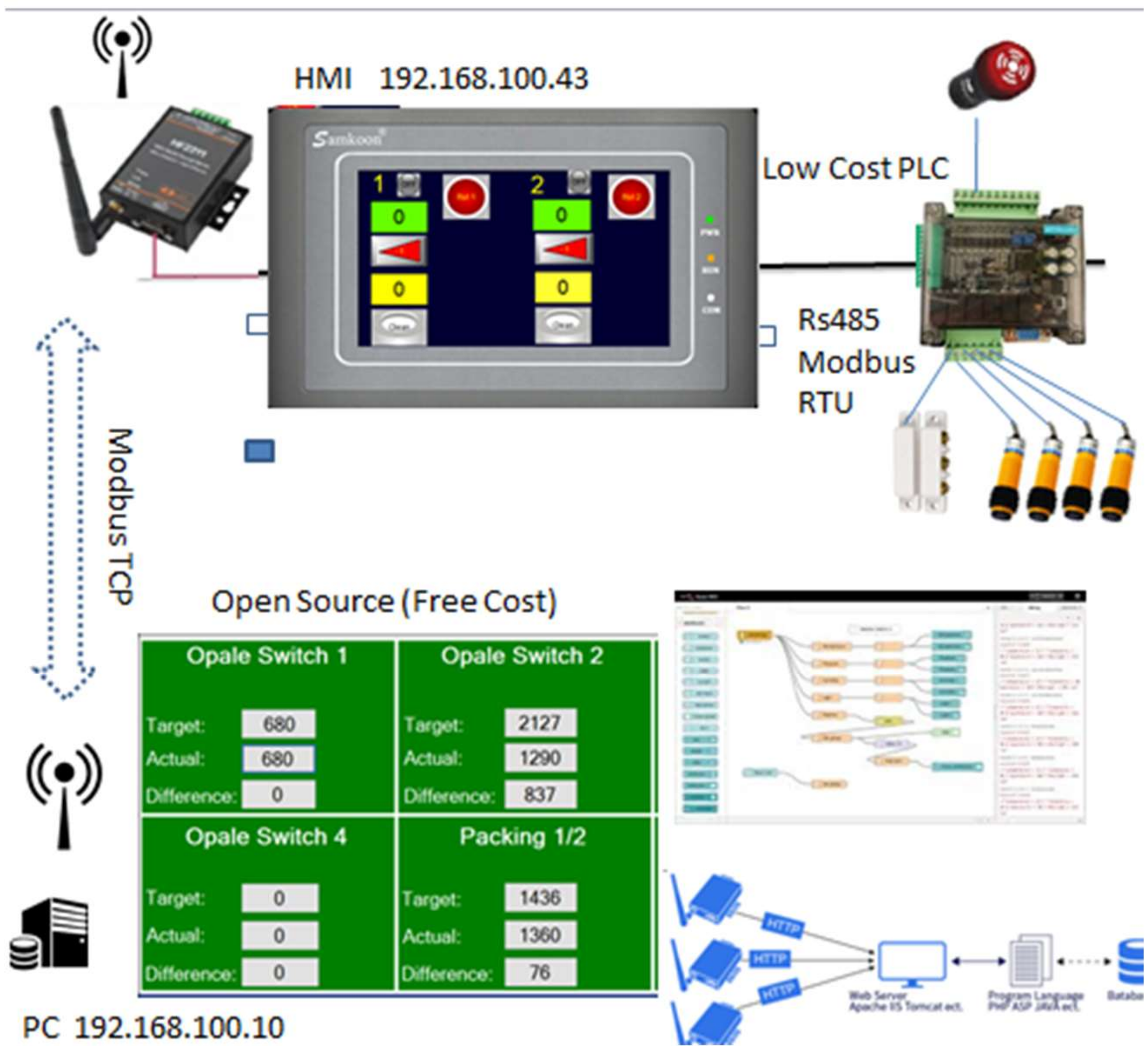

Target:
2127
Actual:
1290
Difference:
837

OMRON E3FA-LP11 Photoelectric Sensor M18, Straight Type, Sensing distance 100mm, Input 10-30VDC

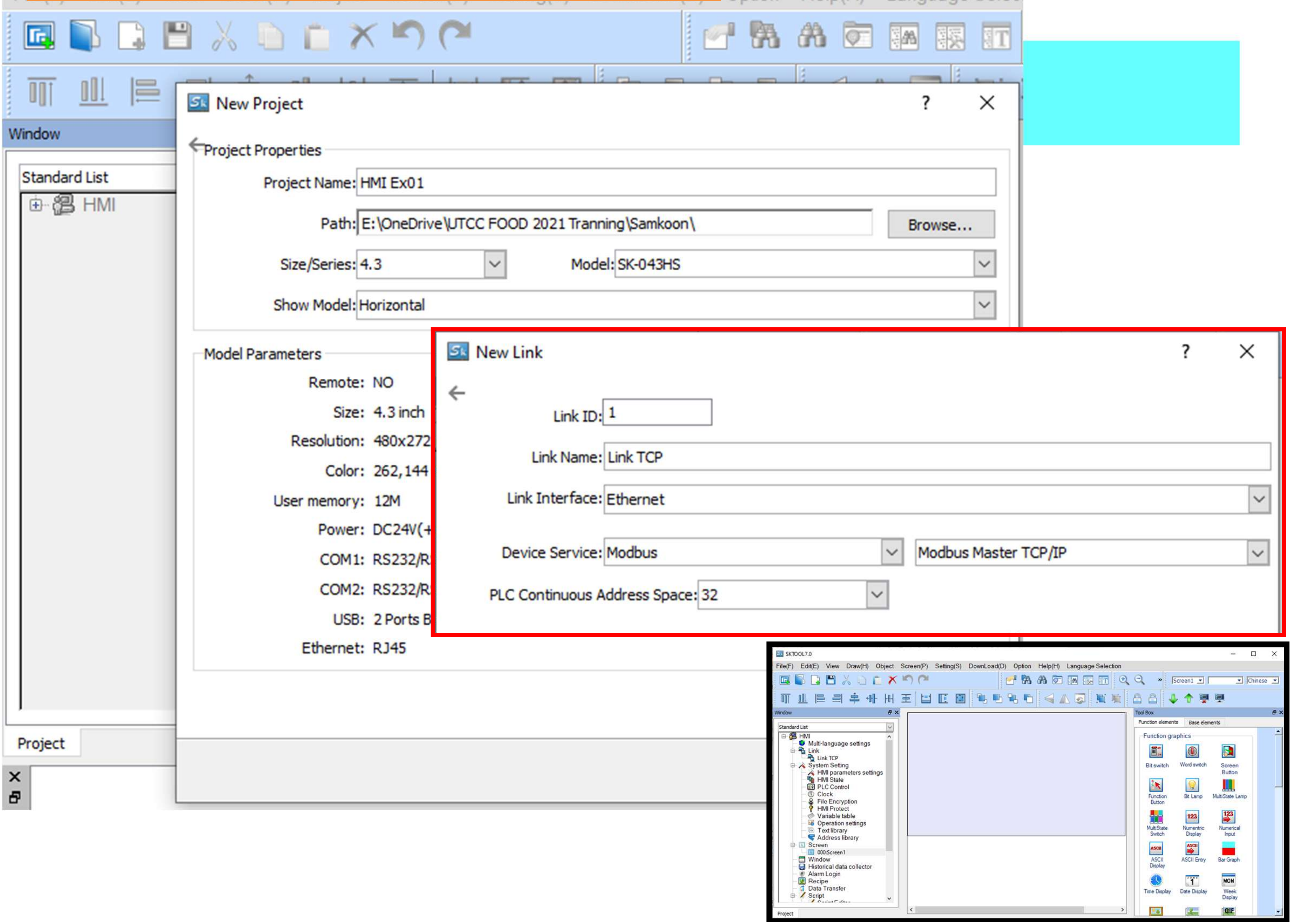



%I0.0
"Proximity"
%Q0.0
"Motor"
%Q0.0
"Motor"
PLC Program
REALPARS

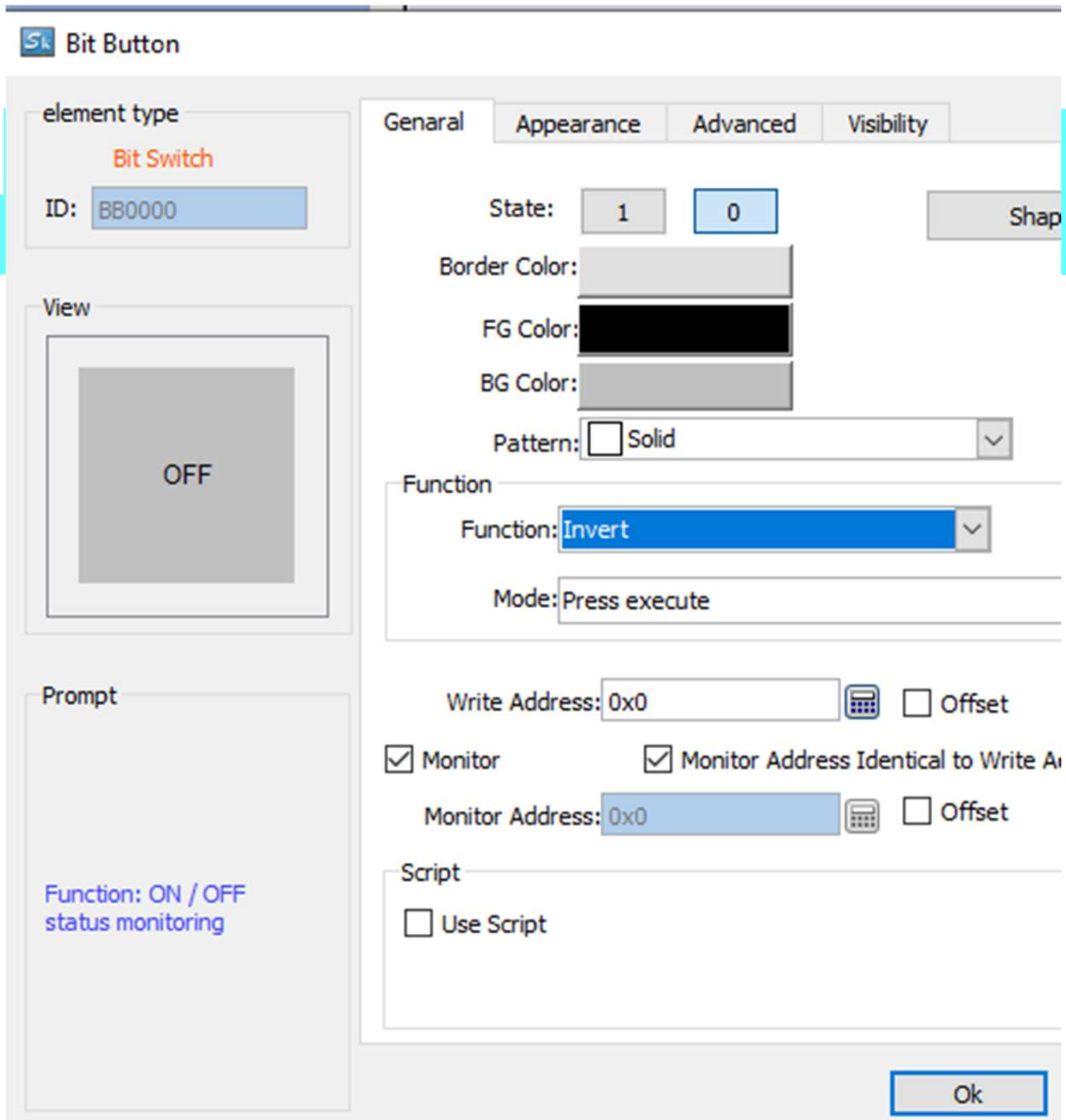
ตัวอย่างการใช้ HMI (Touch Screen) ร่วมกับ PLC

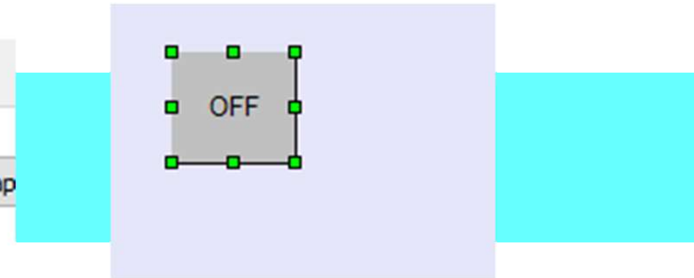


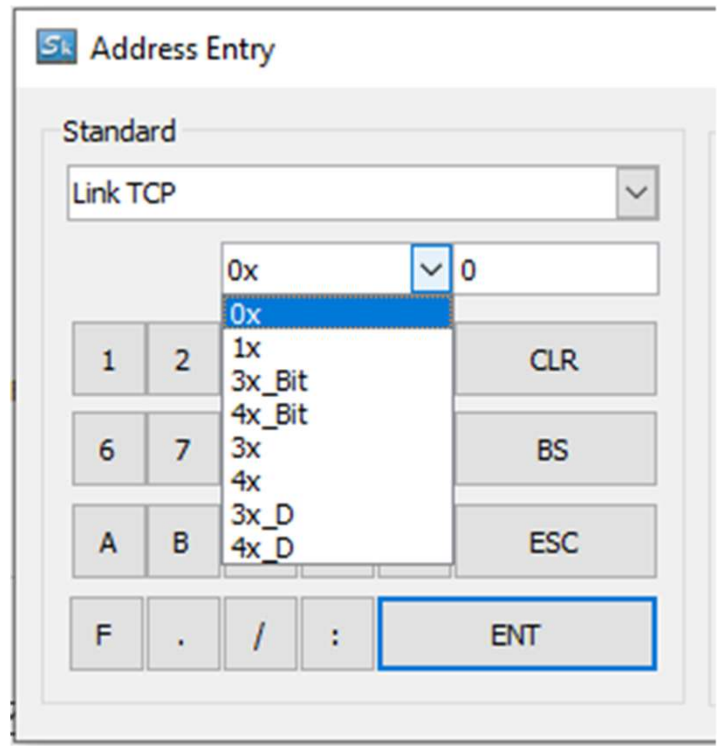
YouTube Modbus – 05 HMI (Samkoon)



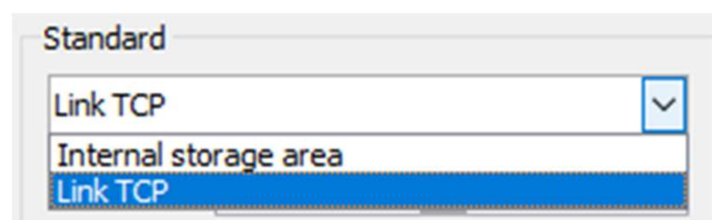

OFF
ON
Bit switch

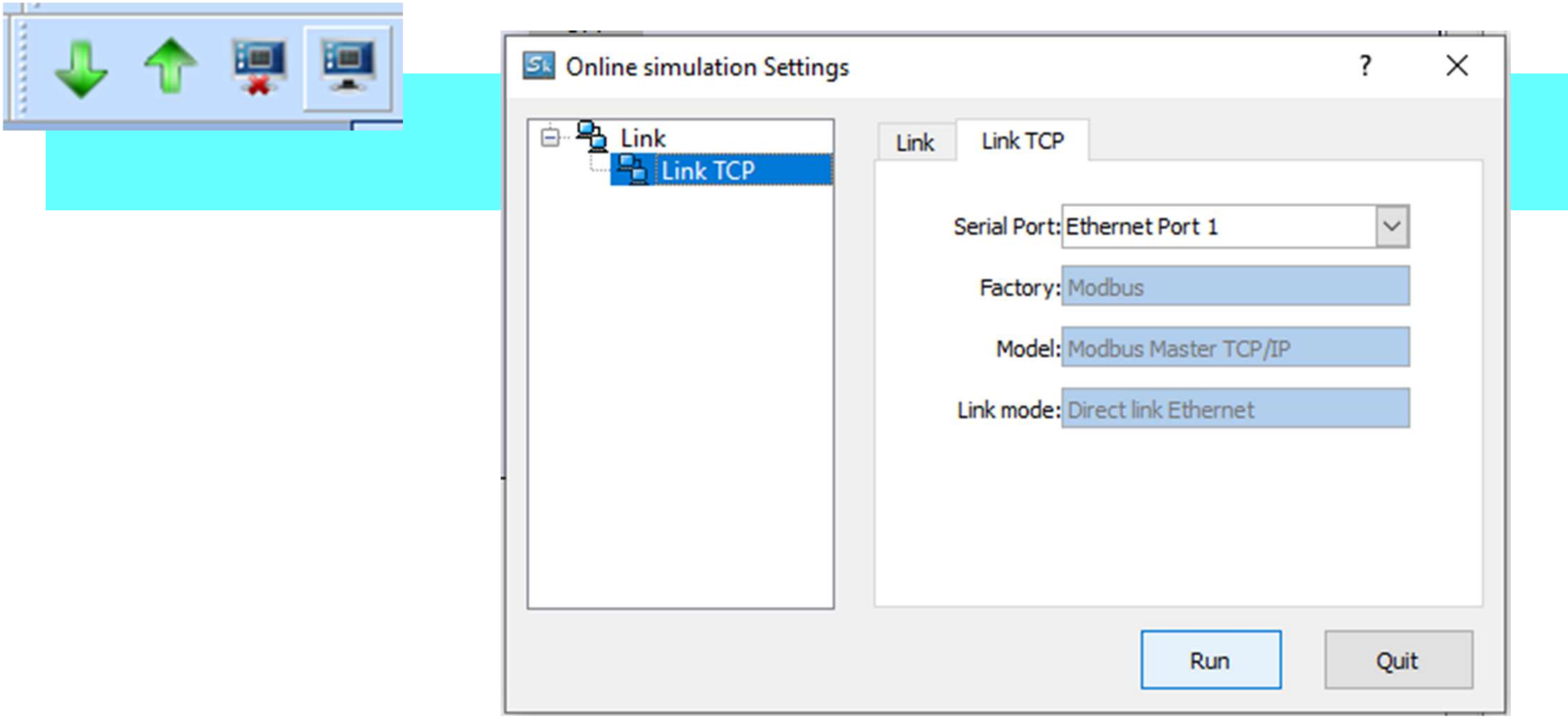

Bit Button
element type
Bit Switch
ID: BB0000
View
OFF
Prompt
Function: ON / OFF
status monitoring
Genaral
Appearance
Advanced
Visibility
State: 1 0
Shap
Border Color:
FG Color:
BG Color:
Pattern: Solid
Function
Function: Invert
Mode: Press execute
Write Address: 0x0
Offset
Monitor
Monitor Address Identical to Write A
Monitor Address: 0x0
Offset
Script
Use Script
Ok

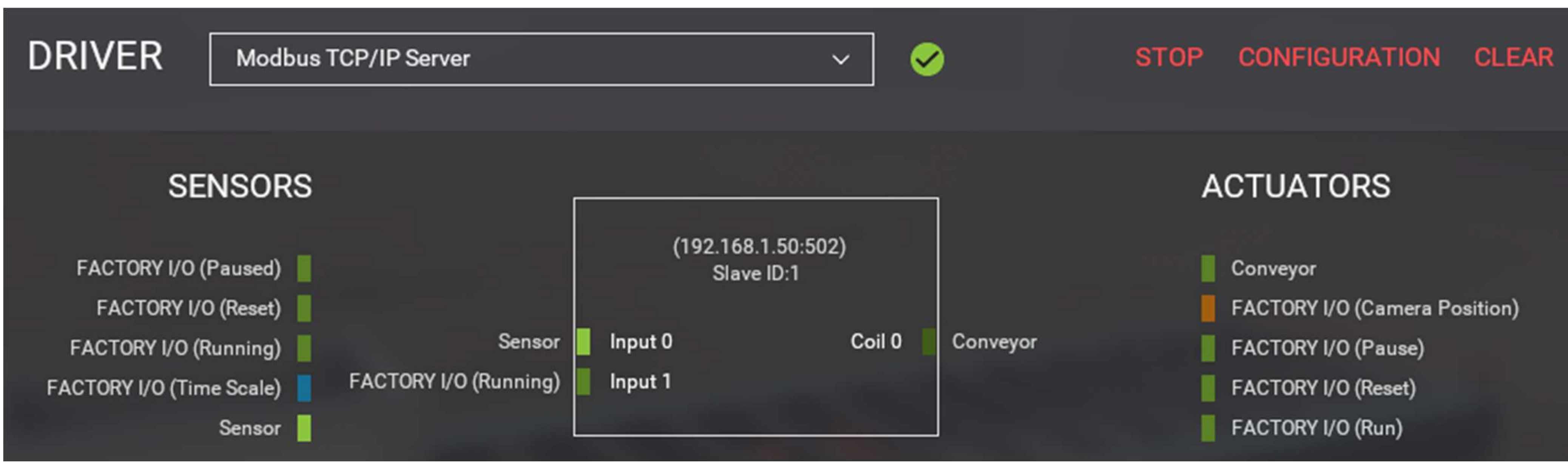

OFF

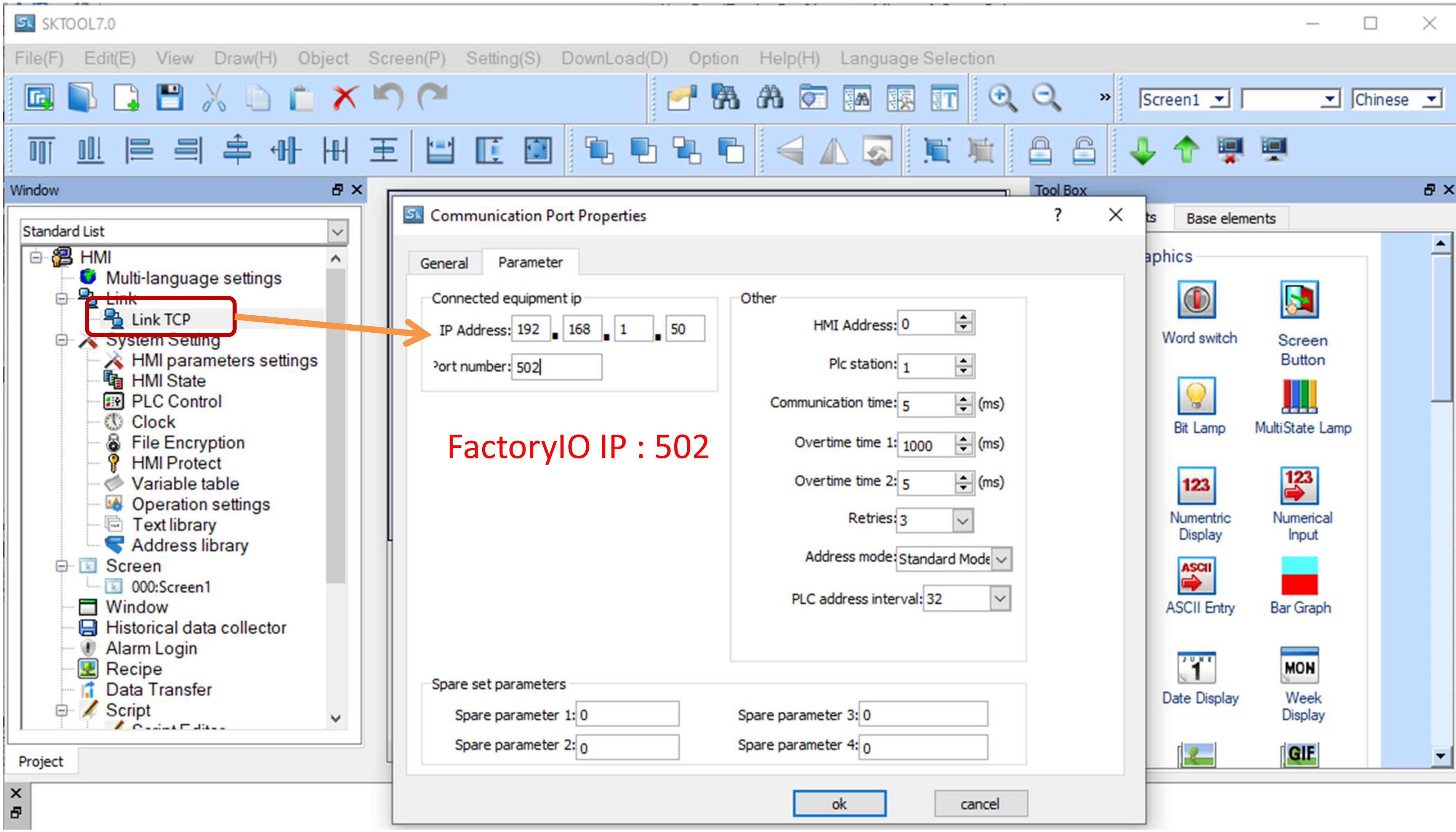

Address Entry
Standard
Link TCP
0x
0
0x
1x
3x_Bit
4x_Bit
3x
4x
3x_D
4x_D
1
2
CLR
6
7
BS
A
B
ESC
F
.
/
:
ENT

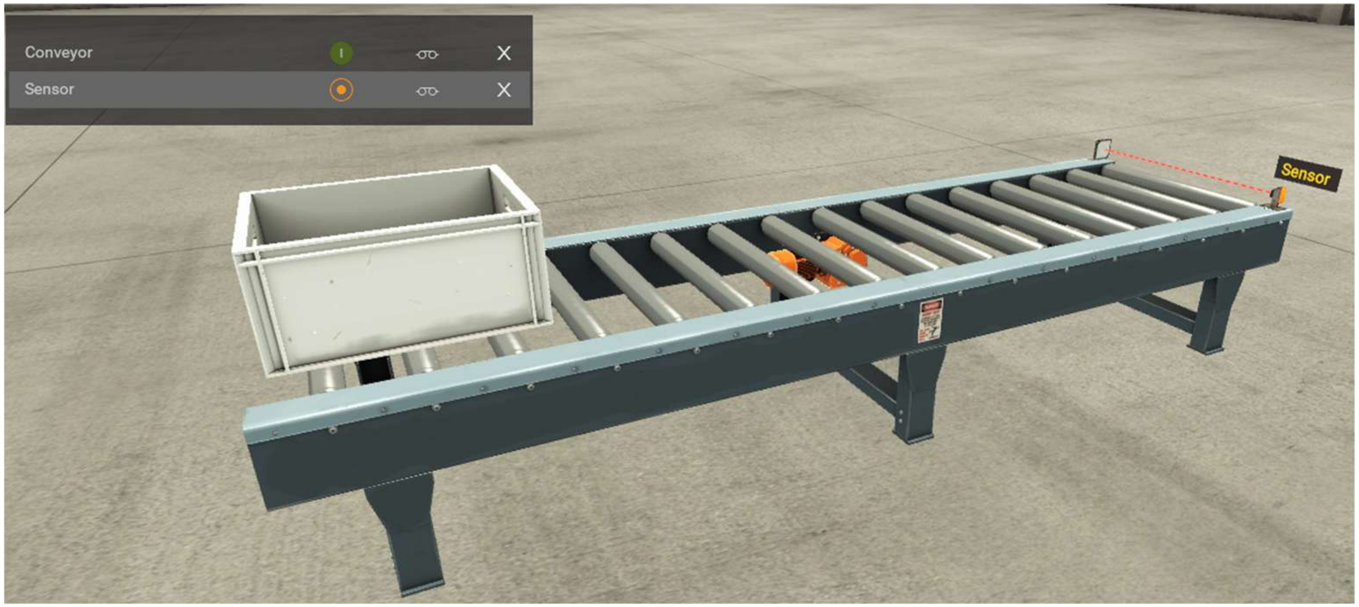

Function
Function: Invert
Mode:
Invert
Inching
Set OFF
Set ON

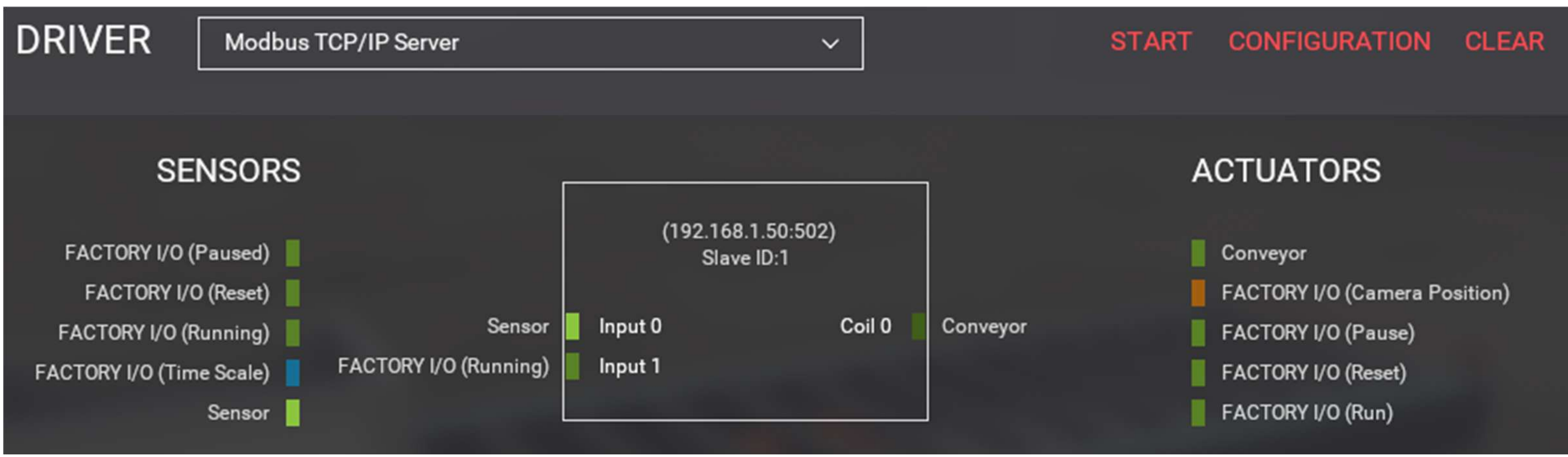

Standard
Link TCP
Internal storage area
Link TCP

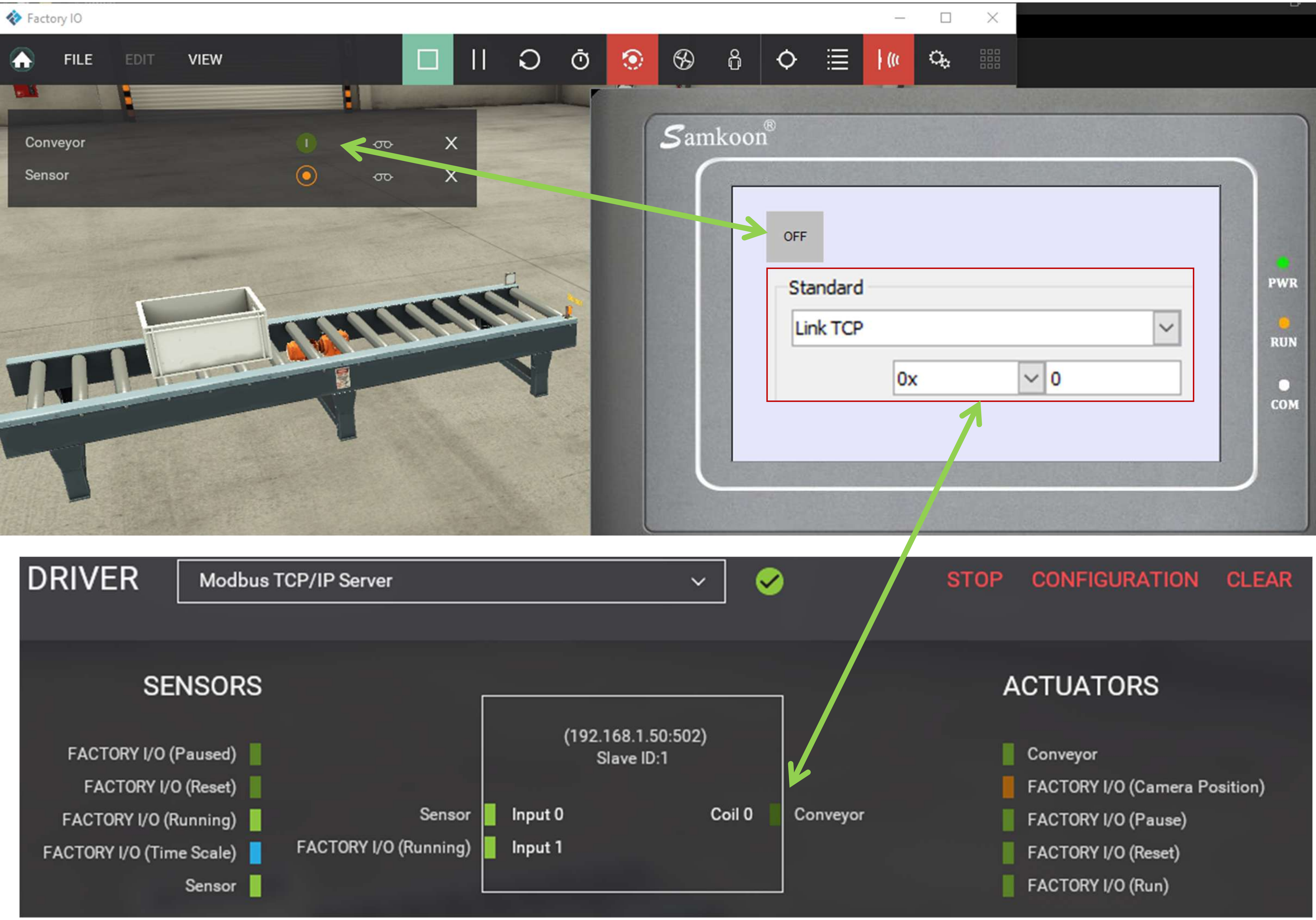

Online simulation Settings
Link
Link TCP
Link
Link TCP
Serial Port: Ethernet Port 1
Factory: Modbus
Model: Modbus Master TCP/IP
Link mode: Direct link Ethernet
Run
Quit


DRIVER
Modbus TCP/IP Server
STOP
CONFIGURATION
CLEAR
SENSORS
ACTUATORS
(192.168.1.50:502)
Slave ID:1
FACTORY I/O (Paused)
FACTORY I/O (Reset)
FACTORY I/O (Running)
FACTORY I/O (Time Scale)
Sensor
Sensor
Input 0
FACTORY I/O (Running)
Input 1
Coil 0
Conveyor
Conveyor
FACTORY I/O (Camera Position)
FACTORY I/O (Pause)
FACTORY I/O (Reset)
FACTORY I/O (Run)

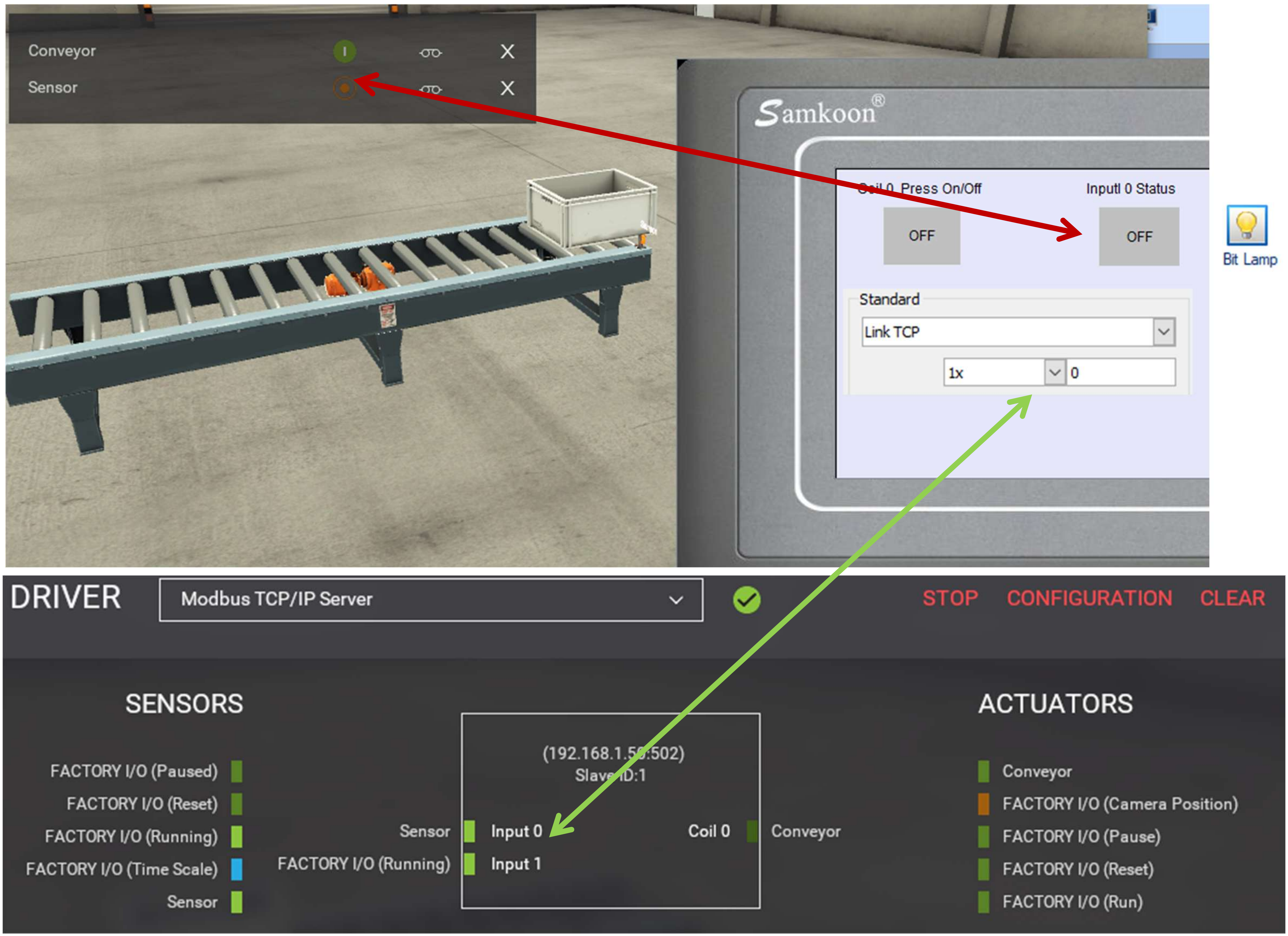

SKTOOL7.0
File(F) Edit(E) View Draw(H) Object Screen(P) Setting(S) DownLoad(D) Option Help(H) Language Selection
Screen1
Chinese
Window
Standard List
HMI
Multi-language settings
Link
Link TCP
System Setting
HMI parameters settings
HMI State
PLC Control
Clock
File Encryption
HMI Protect
Variable table
Operation settings
Text library
Address library
Screen
000:Screen1
Window
Historical data collector
Alarm Login
Recipe
Data Transfer
Script
Project
Communication Port Properties
General
Parameter
Connected equipment ip
IP Address: 192 . 168 . 1 . 50
Port number: 502
FactoryIO IP : 502
Other
HMI Address: 0
Plc station: 1
Communication time: 5 (ms)
Overtime time 1: 1000 (ms)
Overtime time 2: 5 (ms)
Retries: 3
Address mode: Standard Mode
PLC address interval: 32
Spare set parameters
Spare parameter 1: 0
Spare parameter 2: 0
Spare parameter 3: 0
Spare parameter 4: 0
ok
cancel
Tool Box
Base elements
Word switch
Screen Button
Bit Lamp
MultiState Lamp
Numentric Display
Numerical Input
ASCII Entry
Bar Graph
Date Display
Week Display


Conveyor
Sensor
Sensor

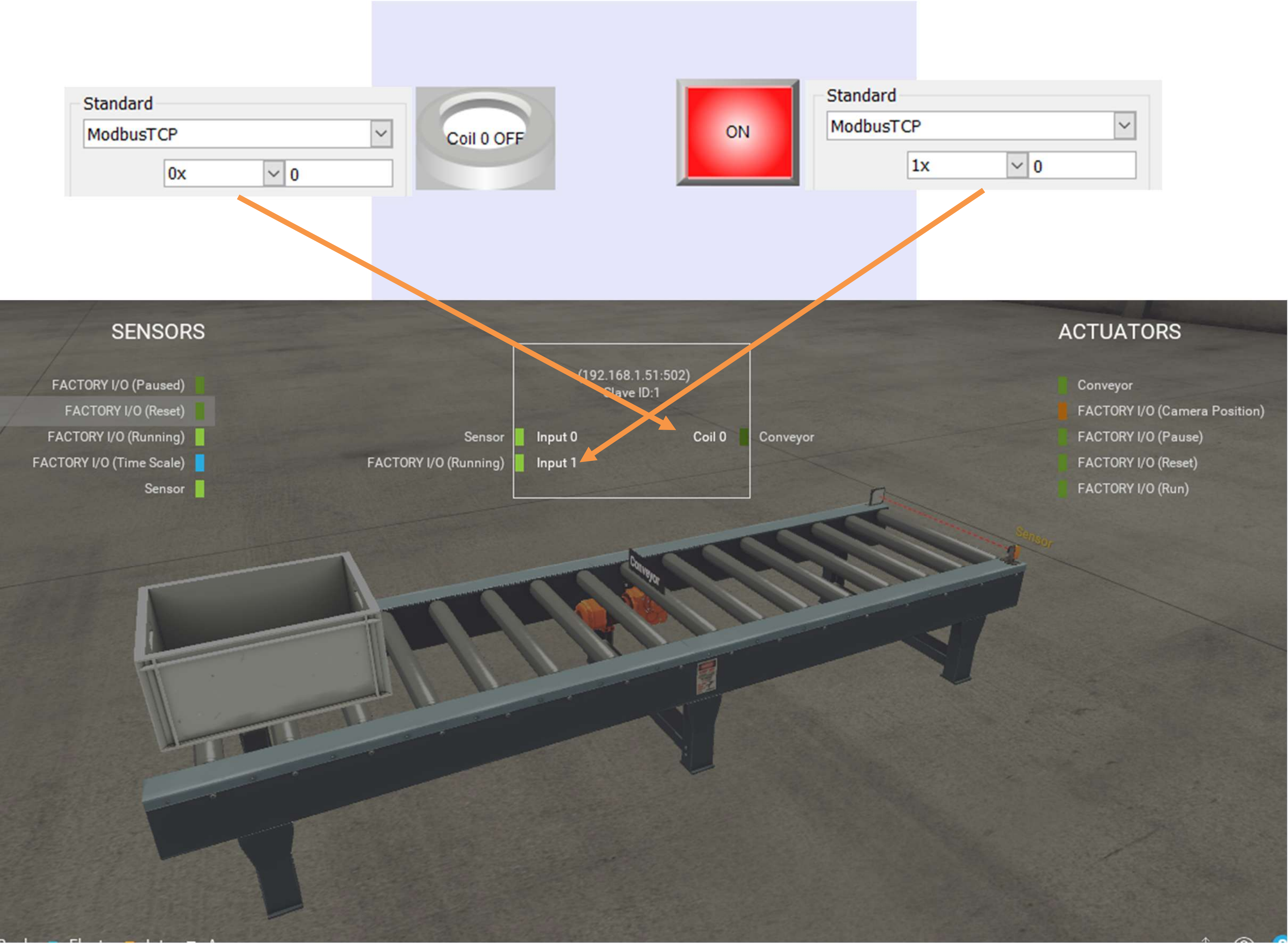

DRIVER
Modbus TCP/IP Server
START
CONFIGURATION
CLEAR
SENSORS
FACTORY I/O (Paused)
FACTORY I/O (Reset)
FACTORY I/O (Running)
FACTORY I/O (Time Scale)
Sensor
(192.168.1.50:502)
Slave ID:1
Sensor
Input 0
FACTORY I/O (Running)
Input 1
Coil 0
Conveyor
ACTUATORS
Conveyor
FACTORY I/O (Camera Position)
FACTORY I/O (Pause)
FACTORY I/O (Reset)
FACTORY I/O (Run)



Factory IO
FILE
EDIT
VIEW
Conveyor
Sensor
Samkoon®
OFF
Standard
Link TCP
0x
0
PWR
RUN
COM
DRIVER
Modbus TCP/IP Server
STOP
CONFIGURATION
CLEAR
SENSORS
FACTORY I/O (Paused)
FACTORY I/O (Reset)
FACTORY I/O (Running)
FACTORY I/O (Time Scale)
Sensor
(192.168.1.50:502)
Slave ID:1
Sensor
Input 0
FACTORY I/O (Running)
Input 1
Coil 0
Conveyor
ACTUATORS
Conveyor
FACTORY I/O (Camera Position)
FACTORY I/O (Pause)
FACTORY I/O (Reset)
FACTORY I/O (Run)

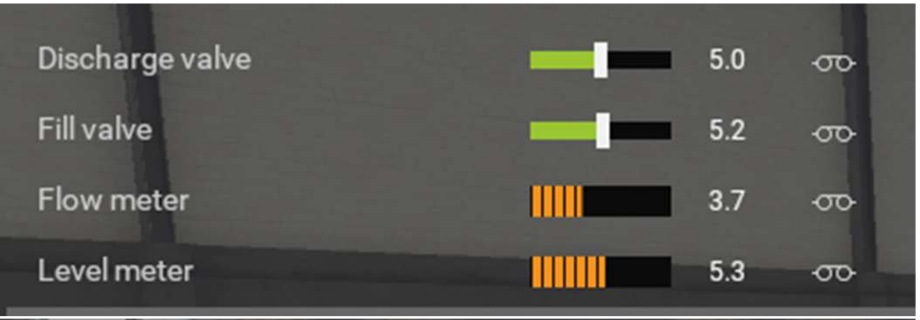


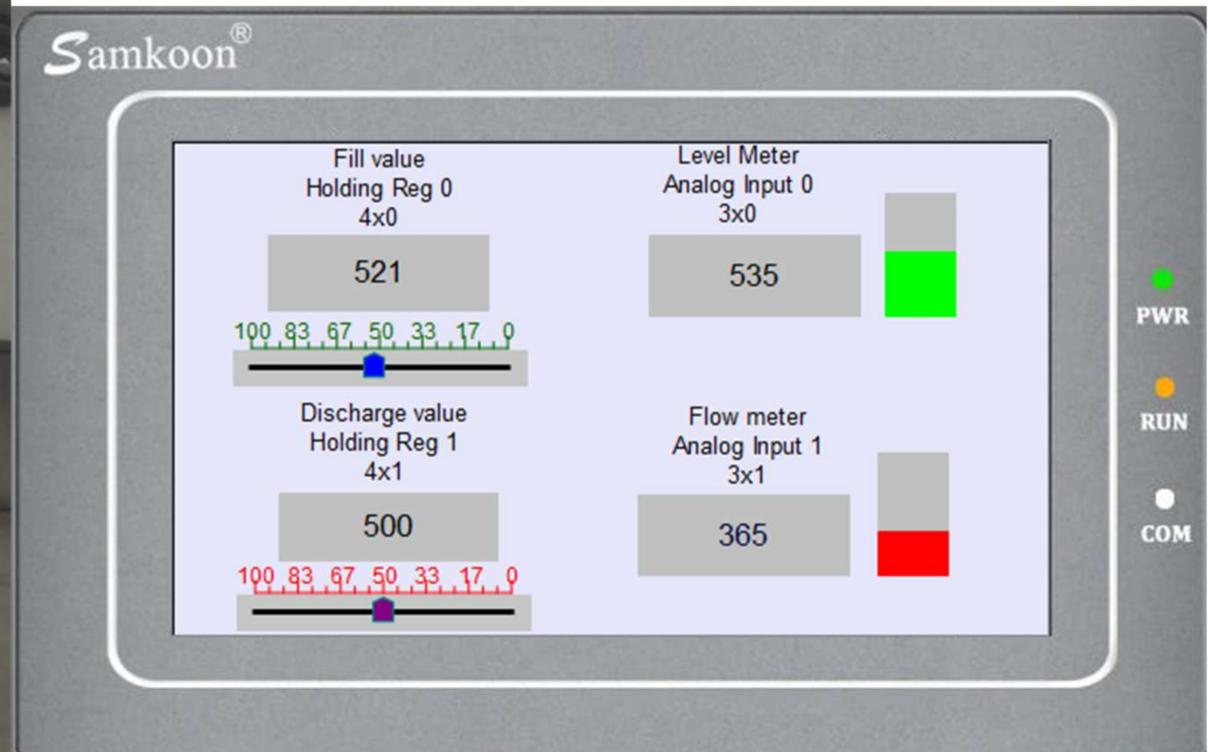
Conveyor
Sensor
Samkoon
Coil 0 Press On/Off
Inputl 0 Status
OFF
OFF
Bit Lamp
Standard
Link TCP
1x
0
DRIVER
Modbus TCP/IP Server
STOP
CONFIGURATION
CLEAR
SENSORS
ACTUATORS
FACTORY I/O (Paused)
FACTORY I/O (Reset)
FACTORY I/O (Running)
FACTORY I/O (Time Scale)
Sensor
(192.168.1.50:502)
Slave ID:1
Sensor
Input 0
FACTORY I/O (Running)
Input 1
Coil 0
Conveyor
Conveyor
FACTORY I/O (Camera Position)
FACTORY I/O (Pause)
FACTORY I/O (Reset)
FACTORY I/O (Run)



Standard
ModbusTCP
0x
0
Coil 0 OFF
ON
Standard
ModbusTCP
1x
0
SENSORS
FACTORY I/O (Paused)
FACTORY I/O (Reset)
FACTORY I/O (Running)
FACTORY I/O (Time Scale)
Sensor
(192.168.1.51:502)
Slave ID:1
Sensor
Input 0
FACTORY I/O (Running)
Input 1
Coil 0
Conveyor
ACTUATORS
Conveyor
FACTORY I/O (Camera Position)
FACTORY I/O (Pause)
FACTORY I/O (Reset)
FACTORY I/O (Run)

YouTube Modbus – 06 HMI Advance





UTCC IIOT (Industrial Internet of Thing) Starter Kid Training

ศูนย์อบรม ช่าง IIOT สำหรับธุรกิจอาหาร Advan IOT,PLC,HMI (ธุรกิจ/โรงงานอาหารสาธิต)

จับเวลา นับสินค้า ควบคุมการผลิต

แสดงสถานะการทำงานปัจจุบัน

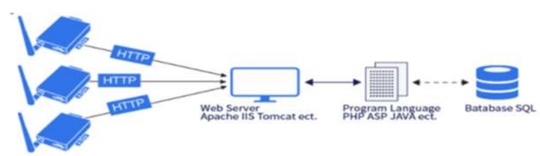
อุณหภูมิ ความชื้น บรรยากาศ

อุณหภูมิภายในเครื่องจักร

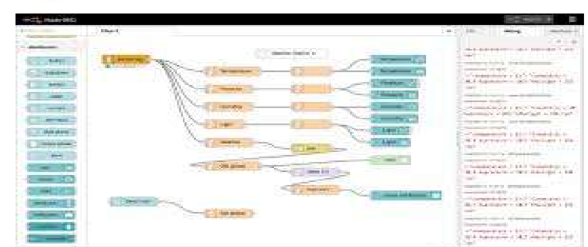
การวัดการใช้พลังงานไฟฟ้า

ควบคุมเครื่องจักรระบบอัตโนมัติ

Modbus RTU

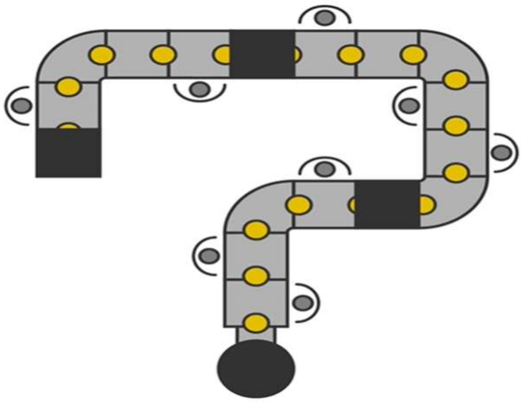


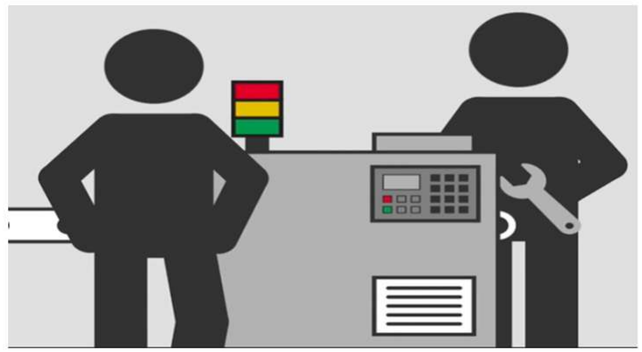
IOT Platform , อบรมซ่อมเครื่องจักร/ ดัดแปลงควบคุมเครื่องจักรด้วย IOT

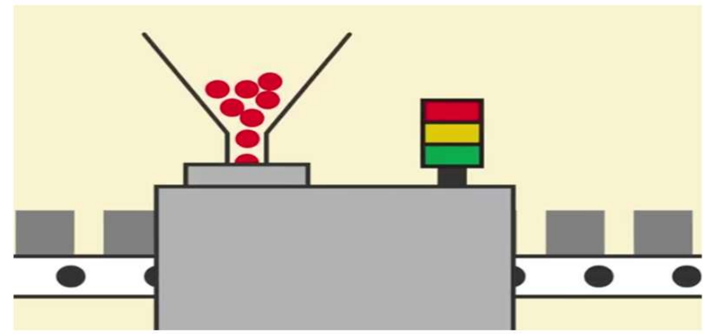


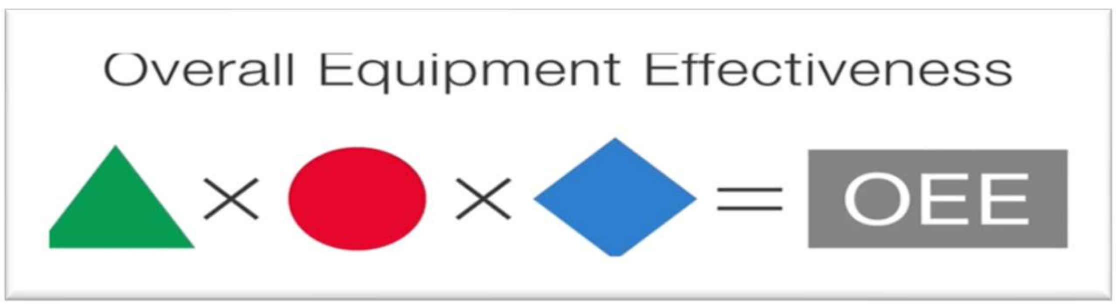
Open Source Software

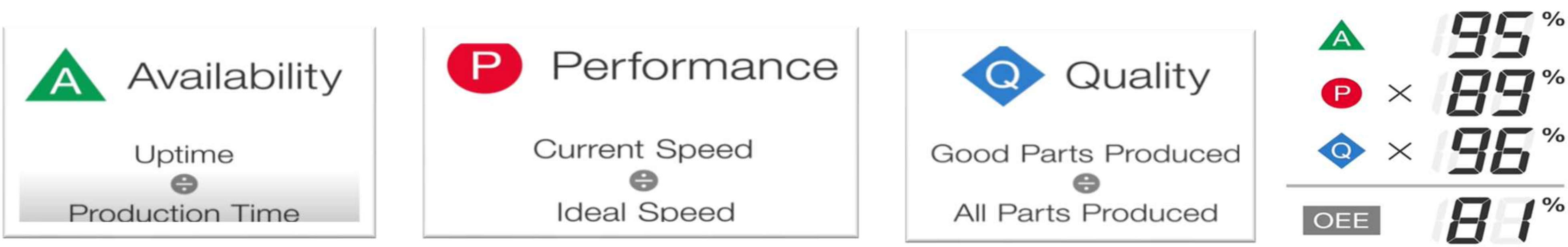
Use OEE to Improve Productivity



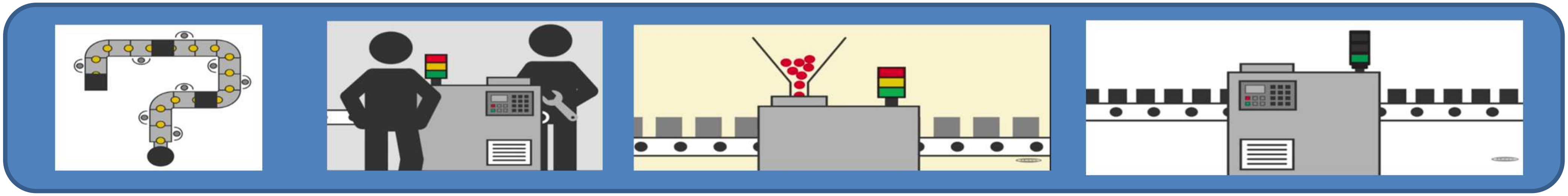


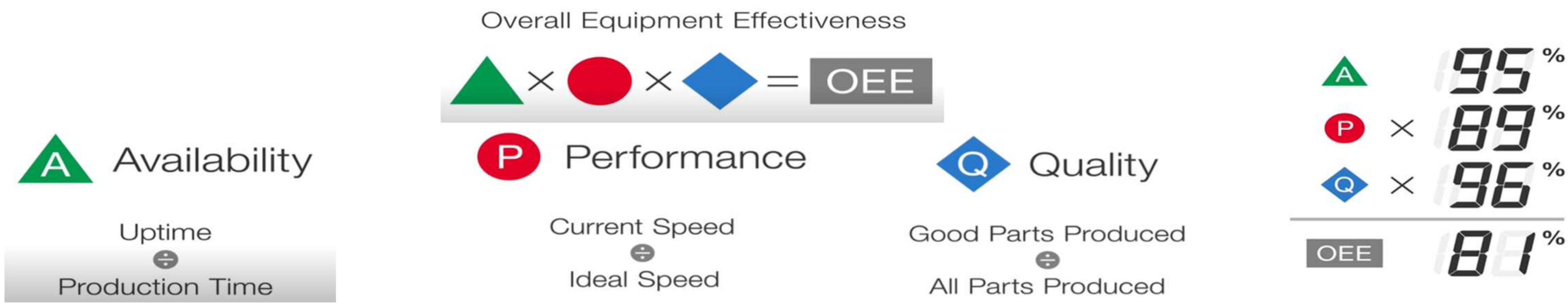


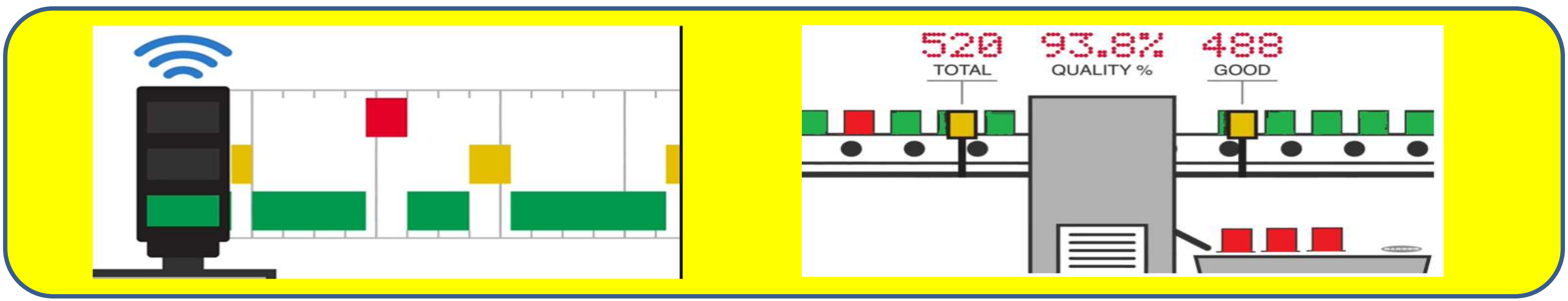


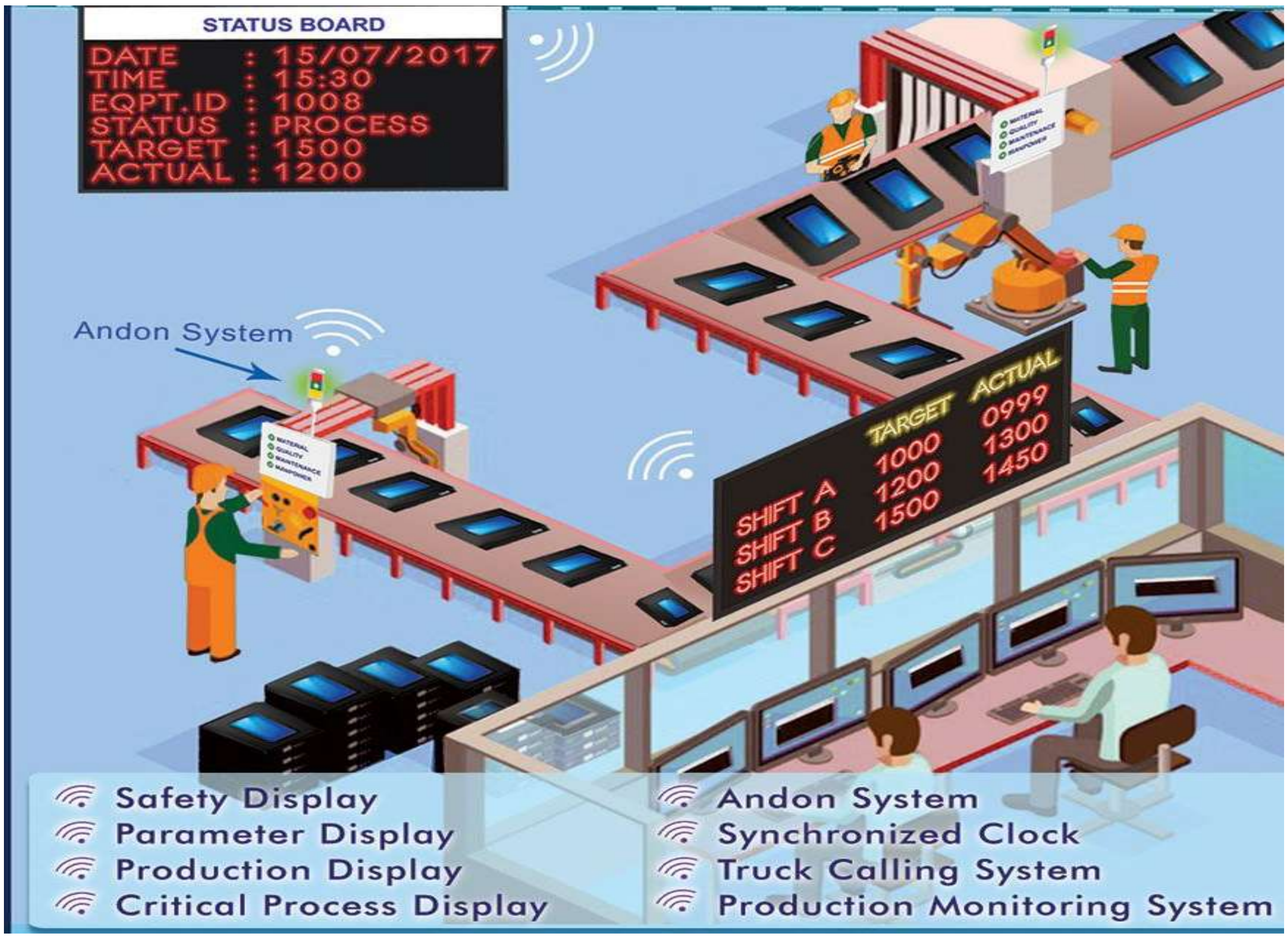


Use OEE to Improve Productivity









Production Monitor

Opale Switch 1		Opale Switch 2	
Target:	680	Target:	2127
Actual:	680	Actual:	1290
Difference:	0	Difference:	837
Opale Switch 4		**Packing 1/2**	
Target:	0	Target:	1436
Actual:	0	Actual:	1360
Difference:	0	Difference:	76